INCH-POUND

MIL-M-28787/436
17 May 1991

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

MEMORY CONTROL MODULE, KEY CODE JMN

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the modules described herein shall consist of this specification and the latest issue of MIL-M-28787.

1. SCOPE

1.1 Scope. This document establishes the requirements and test procedures for the acquisition of an electronic memory control module.

1.2 Classification. This specification covers two classes of modules. Classes I and II as specified in MIL-STD-1389.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Sea Systems Command, Code SEA-55Z3, Department of the Navy, Washington, D.C. 20362-5101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 5963
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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specification and standards. The following specification and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATION

MILITARY

MIL-M-28787 - Modules, Standard Electronic, General Specification for.

STANDARDS

MILITARY

MIL-STD-1389 - Design Requirements for Standard Electronic Modules.

MIL-STD-1665 - Test Equipment for the Standard Electronic Modules Program.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. Modules furnished under this specification shall be products which are authorized by the qualifying activity for listing on the applicable qualified products list by the time of award of contract (see 4.5).

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3.2 General requirements. The module shall comply with the requirements of MIL-M-28787 except as modified herein.

3.3 Correlation requirements. Before any production modules are tested, all testers and test equipment shall be correlated in accordance with MIL-M-28787.

3.4 Failure rate requirements. All modules procured to this specification shall have a predicted failure rate no greater than 1.686 failures per million operating hours for class I or 3.857 failures per million operating hours for class II, when predicted in accordance with MIL-STD-1389. When a class II module is used in a class I environment, the failure rate for class I shall be used for reliability prediction.

3.5 Performance requirements. The module shall conform to the electrical requirements specified in table I and shall operate within the environment of design class I or class II as specified in MIL-STD-1389.

3.5.1 Substitutability. Class II modules are substitutable for class I modules. When class II modules are used in class I environments, the failure rate for class I shall be used for reliability prediction.

3.6 Module configuration requirements. The module configuration requirements shall be in accordance with MIL-STD-1389 and as specified herein.

3.6.1 Key code. The module key code shall be JMN.

3.6.2 Size. The module size shall be 2A Format B.

3.6.3 Weight. The module shall have a maximum weight of 112 grams.

3.6.4 Contact pins. The module shall have 100 contact pins positioned 1 through 50 in one row and 51 through 100 in the other row.

3.6.5 Fin position. The module shall have an insertion and extraction and heat dissipation fin on increment A.

3.6.6 Depth. The module shall have no additional depth requirements.

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TABLE I Module electrical requirements.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
A	Isolation	II	S	10	INF	10	INF	10	INF	MOhm
B1-B6	Continuity	III	X	0	1	0	1	0	1	Ohm
C	Vpp stress	IV	0	----------Module shall pass all remaining tests----------						
D1-D4	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D5	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D6-D9	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D10	Ibd	V	S	0.0	0.6	0.0	0.6	0.0	0.6	mA
D11	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D12	Ibd	V	S	0.0	1.1	0 0	1.1	0 0	1 1	mA
D13-D25	Ibd	V	S	0 0	0.1	0 0	0.1	0 0	0 1	mA
D26	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D27	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D28	Ibd	V	S	0.0	1.4	0.0	1.4	0.0	1.4	mA
D29	Ibd	V	S	0.0	0.1	0 0	0.1	0 0	0 1	mA
D30	Ibd	V	S	0.0	0.3	0.0	0.3	0 0	0 3	mA
D31,D32	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D33	Ibd	V	S	-0.58	1.67	-0.58	1.67	-0.58	1.67	mA
D34	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D35	Ibd	V	S	-0.58	0.78	-0.58	0.78	-0.58	0.78	mA
D36-D38	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0.1	mA
D39	Ibd	V	S	0.0	0.2	0.0	0.2	0.0	0.2	mA
D40	Ibd	V	S	0.0	0.1	0.0	0.1	0.0	0 1	mA
D41	Ibd	V	S	-0.12	0.22	-0.12	0.22	-0.12	0.22	mA
D42	Ibd	V	S	-0.58	0.68	-0.58	0.68	-0 58	0 68	mA
D43	Ibd	V	S	-0.58	1.67	-0.58	1.67	-0 58	1 67	mA
E1	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E2	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E3	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E4	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E5	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
E6	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E7	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E8,E9	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E10	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E11	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E12	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E13	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E14	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E15	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E16,E17	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E18	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E19	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E20	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E21	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E22	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E23	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E24,E25	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E26	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E27	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E28	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E29	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E30	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E31	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E32-E36	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E37	Ios	VI	S	-325.0	-100.0	-325.0	-100.0	-325.0	-100.0	mA
E38-E43	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E44,E45	Ios	VI	S	-650.0	-200.0	-650.0	-200.0	-650.0	-200.0	mA
E46	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
E47	Ios	VI	S	-350.0	-60.0	-350.0	-60.0	-350.0	-60.0	mA
E48-E59	Ios	VI	S	-150.0	-60.0	-150.0	-60.0	-150.0	-60.0	mA
F1-F4	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F5	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F6-F9	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
F10	Iih	VII	X	-3.6	0.12	-3.6	0.12	-3.6	0.12	mA
F11	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F12	Iih	VII	X	-2.2	0.07	-2.2	0.07	-2 2	0.07	mA
F13-F25	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F26	Iih	VII	X	-6.03	-2.85	-6.03	-2.85	-6.03	-2.85	mA
F27	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F28	Iih	VII	X	-10.6	0.18	-10.6	0.18	-10.6	0.18	mA
F29	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F30	Iih	VII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
F31,F32	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F33,F34	Iih	VII	X	-29.7	-14.3	-29.7	-14.3	-29.7	-14.3	mA
F35	Iih	VII	X	-1.2	0.04	-1 2	0.04	-1.2	0.04	mA
F36	Iih	VII	X	-28.3	-14.3	-28.3	-14.3	-28 3	-14.3	mA
F37-F39	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F40	Iih	VII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
F41	Iih	VII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
F42	Iih	VII	X	-6.03	-2.85	-6.03	-2.85	-6.03	-2.85	mA
F43	Iih	VII	X	-27.7	-14.3	-27.7	-14.3	-27 7	-14.3	mA
F44	Iih	VII	X	-29.7	-14.3	-29.7	-14.3	-29.7	-14.3	mA
G1-G4	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G5	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G6-G9	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G10	Iil	VIII	X	-3.6	0.12	-3.6	0.12	-3.6	0.12	mA
G11	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G12	Iil	VIII	X	-2.2	0.07	-2.2	0.07	-2.2	0.07	mA
G13-G25	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G26	Iil	VIII	X	-6.03	-2.85	-6.03	-2.85	-6.03	-2.85	mA
G27	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G28	Iil	VIII	X	-10.6	0.18	-10.6	0.18	-10.6	0.18	mA
G29	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G30	Iil	VIII	X	-1.8	0.06	-1.8	0.06	-1.8	0.06	mA
G31,G32	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G33,G34	Iil	VIII	X	-29.7	-14.3	-29.7	-14.3	-29.7	-14.3	mA

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
G35	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G36	Iil	VIII	X	-28.3	-14.3	-28.3	-14.3	-28.3	-14.3	mA
G37-G39	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G40	Iil	VIII	X	-1.2	0.04	-1.2	0.04	-1.2	0.04	mA
G41	Iil	VIII	X	-0.6	0.02	-0.6	0.02	-0.6	0.02	mA
G42	Iil	VIII	X	-6.03	-2.85	-6.03	-2.85	-6.03	-2.85	mA
G43	Iil	VIII	X	-27.7	-14.3	-27.7	-14.3	-27.7	-14.3	mA
G44	Iil	VIII	X	-29.7	-14.3	-29.7	-14.3	-29.7	-14.3	mA
H1-H60	Voh	IX	X	2.4	4.5	2.4	4.5	2.4	4.5	V
J1-J63	Vol	X	X	0.0	0.5	0.0	0.5	0.0	0.5	V
K1	Iozh	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
K2	Iozh	XI	X	0.0	0.25	0.0	0.25	0.0	0.25	mA
K3	Iozh	XI	X	0.0	0.5	0.0	0.5	0.0	0.5	mA
K4	Iozl	XI	X	-0.05	0.05	-0.05	0.05	-0.05	0.05	mA
L1-L37	Vic	XII	S	-1.2	0.0	-1.2	0.0	-1.2	0.0	V
M1	PSFC	XIII	S	3.6	6.8	3.6	6.8	3.6	6.8	uF
M2	PSFC	XIII	S	2.1	3.9	2.1	3.9	2.1	3.9	uF
N	Icc	XIV	X	850	1560	850	1560	650	1560	mA
P1,P2	Functional	XV	X	----------			See table XV			
Q1,Q2	Dynamic	XVI	S	----------			See table XVI			
Q3,Q4	Dynamic	XVI	Q	----------			See table XVI			
R1-R4	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R5	Ci/o	XVIII	Q	4	100	4	100	4	100	pF
R6-R9	Ci/o	XVIII	Q	2	75	2	75	2	75	pF
R10	Ci/o	XVIII	Q	12	100	12	100	12	100	pF

- See footnotes at end of table.

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Test cond	Parameter 5/	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/ 0 ± 5°C 60 ± 5°C		Class II 4/ -55 ± 5°C 85 ± 5°C		Units
				Min	Max	Min	Max	Min	Max	
R11	Ci/D	XVIII	Q	2	50	2	50	2	50	pF
R12	Ci/D	XVIII	Q	4	75	4	75	4	75	pF
R13	Ci/D	XVIII	Q	4	50	4	50	4	50	pF
R14-R25	Ci/D	XVIII	Q	2	50	2	50	2	50	pF
R26	Ci/D	XVIII	Q	4	75	4	75	4	75	pF
R27	Ci/D	XVIII	Q	2	50	2	50	2	50	pF
R28	Ci/D	XVIII	Q	4	200	4	200	4	200	pF
R29	Ci/D	XVIII	Q	2	75	2	75	2	75	pF
R30	Ci/D	XVIII	Q	6	100	6	100	6	100	pF
R31,R32	Ci/D	XVIII	Q	2	75	2	75	2	75	pF
R33	Ci/D	XVIII	Q	2	200	2	200	2	200	pF
R34	Ci/D	XVIII	Q	6	100	6	100	6	100	pF
R35	Ci/D	XVIII	Q	4	100	4	100	4	100	pF
R36	Ci/D	XVIII	Q	6	75	6	75	6	75	pF
R37,R38	Ci/D	XVIII	Q	2	50	2	50	2	50	pF
R39	Ci/D	XVIII	Q	2	216	2	216	2	216	pF
R40	Ci/D	XVIII	Q	4	100	4	100	4	100	pF
R41	Ci/D	XVIII	Q	2	50	2	50	2	50	pF
R42,R43	Ci/D	XVIII	Q	4	75	4	75	4	75	pF
R44	Ci/D	XVIII	Q	2	175	2	175	2	175	pF
T1,T2	PSTR	XIX	Q	0	400	0	400	0	400	mV

- See footnotes on next page.

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1/ "X" parameters shall be 100 percent tested during the module acceptance test.

2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.

3/ "Q" parameters shall be tested during qualification testing only.

4/ Unless otherwise specified in the test table, limits are for initial and end-of-life.

5/ Abbreviations used in table I are defined in 6.1.

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3.6.7 Holddown holes. The module shall have no additional requirements for location or quantity of holddown holes.

3.6.8 Marking. Marking shall be as specified in MIL-STD-1389, and the module shall be marked with the military part or identifying number M28787/436-1 for class I and M28787/436-2 for class II. No marking shall interfere with heat dissipation from the fin (such as affixing a label which may act as a heat barrier).

3.7 Circuit pin configuration. The functional block diagram shall conform to figure 1. The pin assignments shall conform to figure 2.

3.8 Acceptance test. All production modules shall be tested in accordance with footnotes 1 and 2 in table I.

3.9 End-of-life limits. Modules which have been plugged into an operating system, or exceed 2 years from date of manufacture and are to be tested, shall be capable of conforming to the 25°C end-of-life limits in table I (see 6.2).

3.10 Power supply voltage. Unless otherwise specified herein, the module shall be capable of conforming to the requirements in table I for a power supply voltage (Vcc) of 5.5 V dc.

3.11 Parallel outputs. Paralleling of outputs is not permissible.

4. QUALITY ASSURANCE PROVISIONS

4.1 Quality assurance. The quality assurance provisions of MIL-M-28787 shall apply.

4.2 Acceptance tests. Acceptance tests of all modules shall be in accordance with procedures and equipment specified in MIL-M-28787.

4.3 Test equipment. Test equipment specified on figures and in tables with a designator shall be as specified in MIL-STD-1665. Other equipment and components shall be as specified on the figures. Unless otherwise specified herein, all resistors shall be metal film and all capacitors shall be ceramic.

4.4 Test procedures. The module shall be tested as specified in table I.

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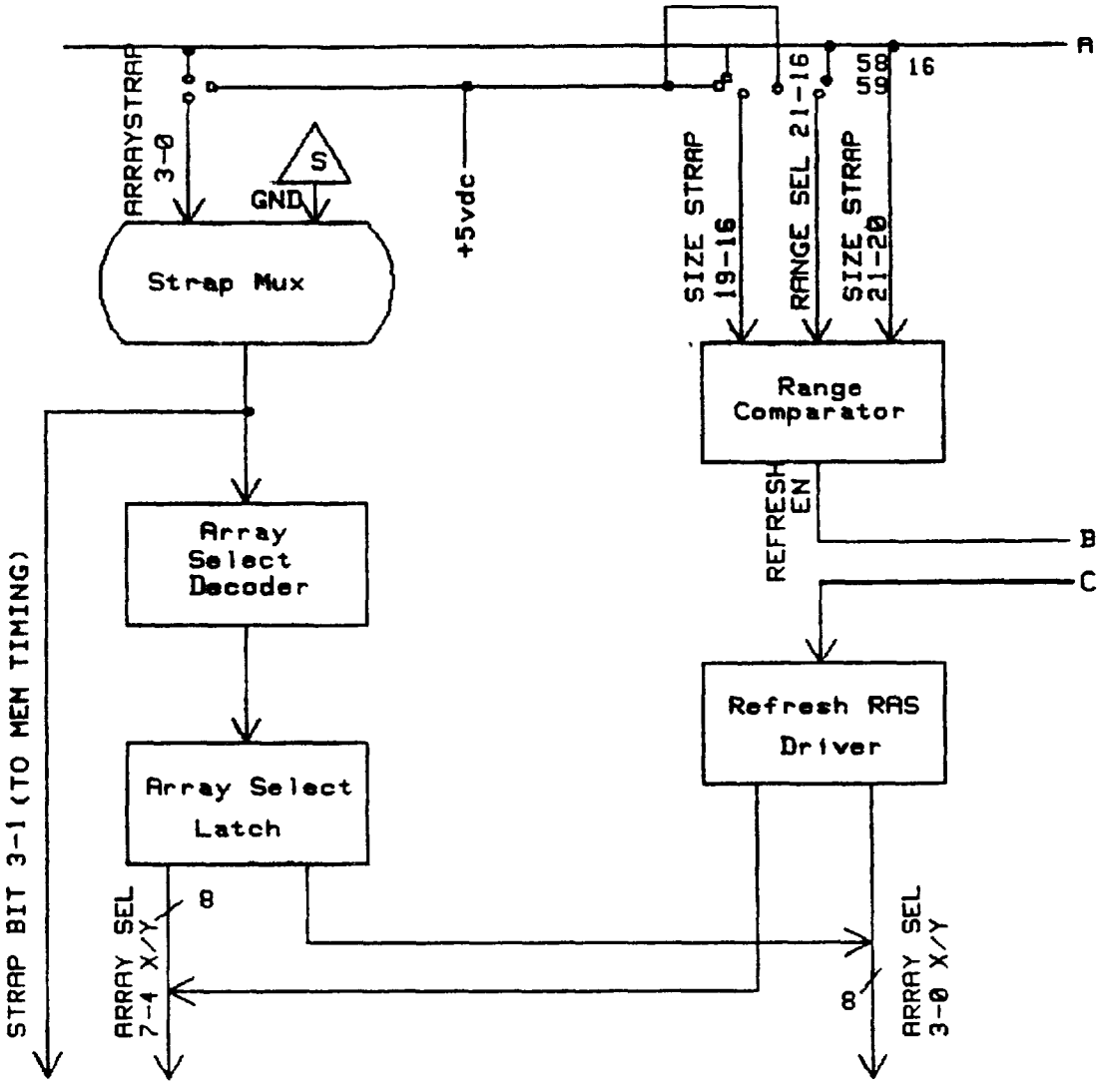


FIGURE 1a. JMN functional block diagram.

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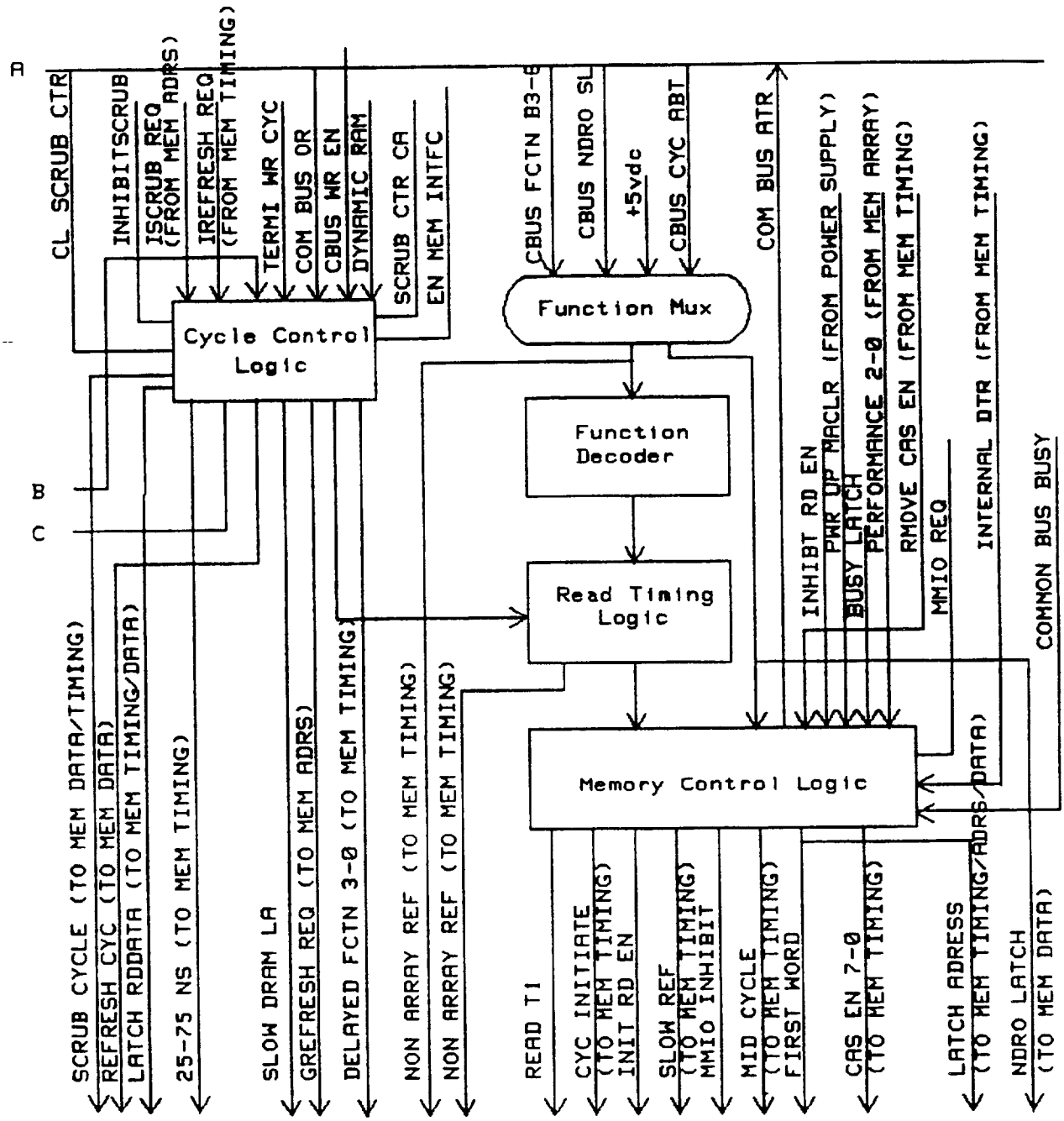


FIGURE 1b. JMN functional block diagram - Continued.

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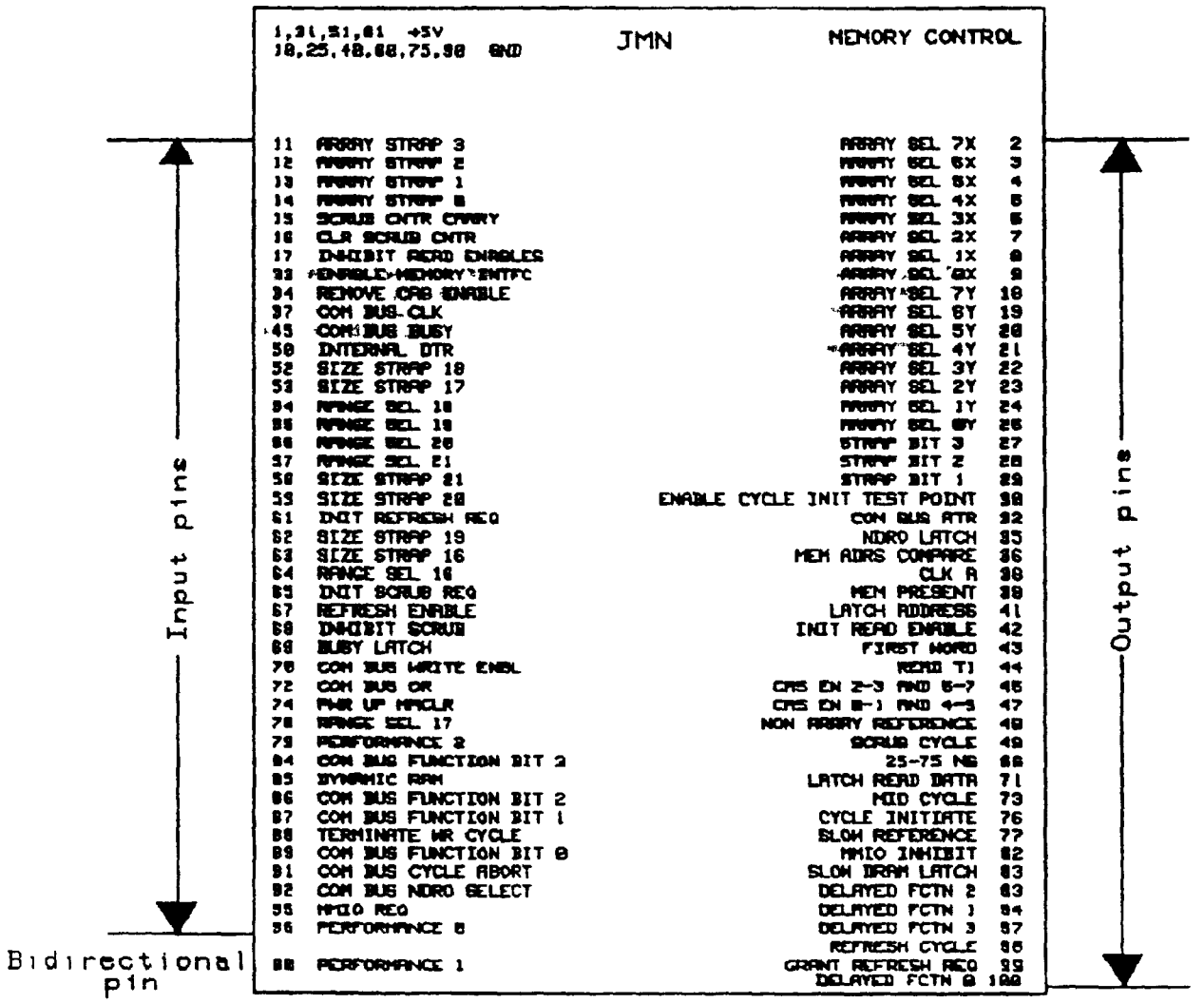


FIGURE 2. JMN pin assignment.

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4.4.1 Alternate test methods. Alternate test methods may be employed if approved by the Standard Electronic Modules Quality Assurance Activity (SEM-QAA) and they conform to the requirements of MIL-M-28787.

4.4.2 Order of testing. Test conditions A through T shall be performed in any order with the following exceptions. Test conditions A, C, D, E, and L are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, C, D, E or L are part of a sequence of tests, the order of testing shall be as follows: A, C, D, E, and L where applicable, followed by the remaining tests in the sequence.

4.4.3 Multiple testing of pins. Singular pins listed multiple times in a test table indicate multiple devices connected to the pins. The module shall be tested so that each device connected to a pin is exercised and tested.

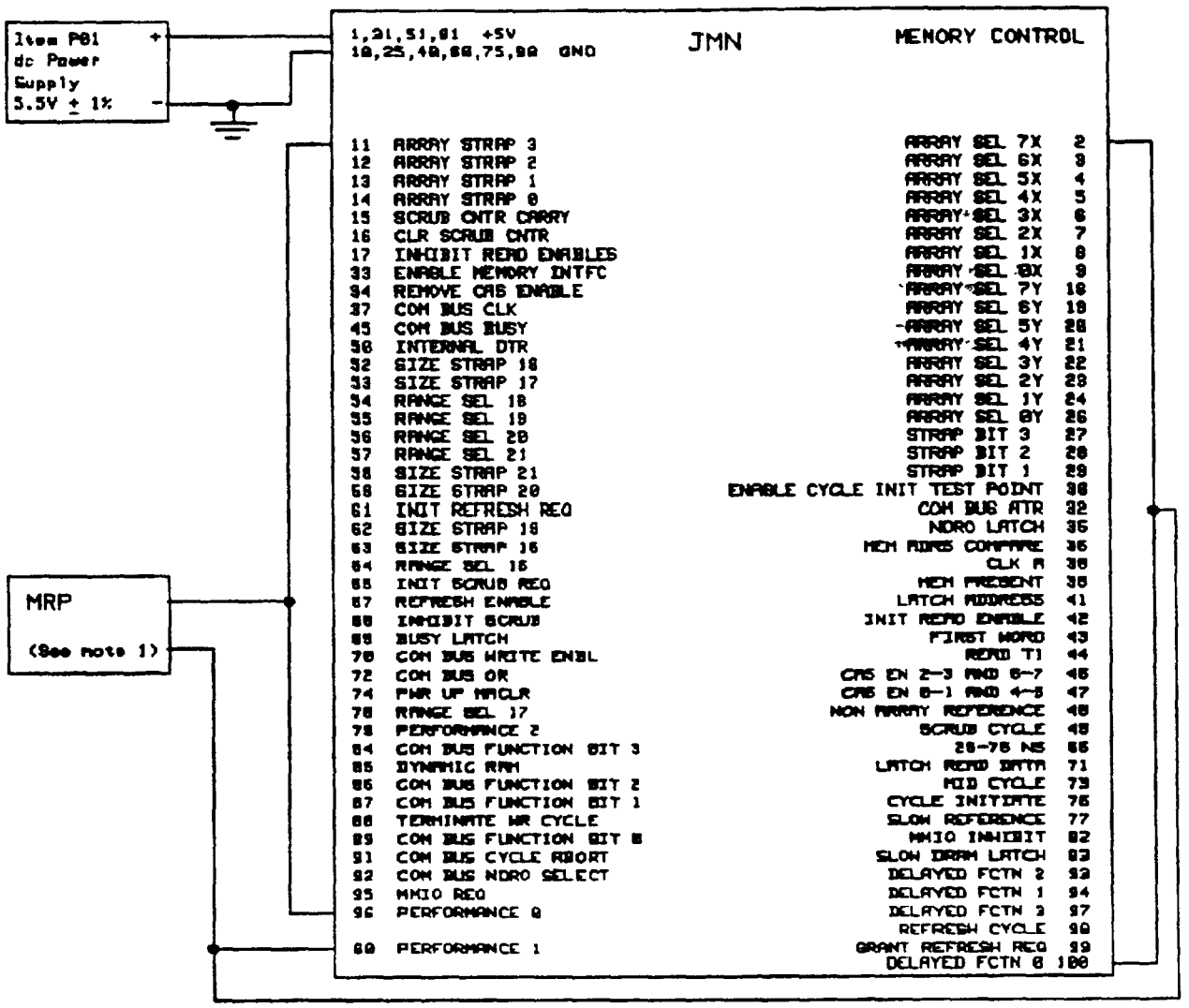
4.5 Qualification testing. Initial qualification tests and periodic check tests shall be performed in accordance with the test plan specified in MIL-M-28787 with the following modifications or additions.

4.5.1 Operating temperature tests. The operating temperature tests shall be conducted in accordance with MIL-M-28787. Before starting these tests, test conditions A through T in table I shall be performed in the order specified in 4.4.2 using the initial limits at 25°C ambient. When electrical testing is called for at the end of each temperature stabilization period, test conditions B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the initial limits for the applicable operating temperature. At all other times during the operating temperature tests, the module shall be operated in the test circuit shown on figure 3.

4.5.1.1 Component temperature test. The component temperature test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 3.

4.5.2 Vibration tests. The vibration tests shall be conducted in accordance with MIL-M-28787. During the vibration tests, the module shall be operated in the test circuit shown on figure 4. Following the vibration tests, the electrical tests in table I which are marked with an "X" in the test type column, and test condition M, shall be performed using the initial limits at 25 ± 5°C ambient.

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NOTE:

1. Military Reconfigurable Processor (MRP) or emulator capable of reproducing system backpanel characteristics.

FIGURE 3. JMN operating temperature and component temperature test circuit.

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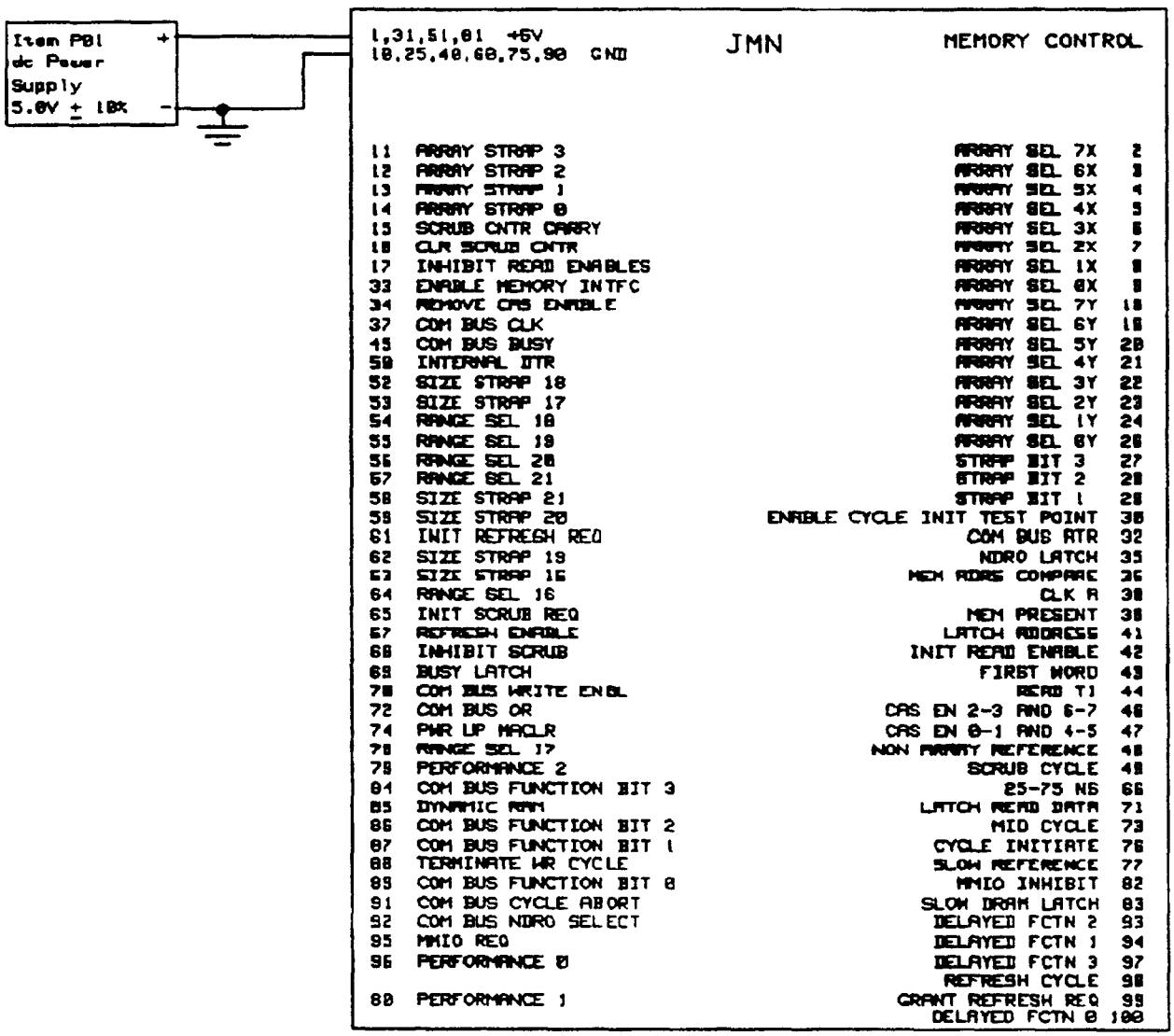


FIGURE 4. JMN vibration test circuit.

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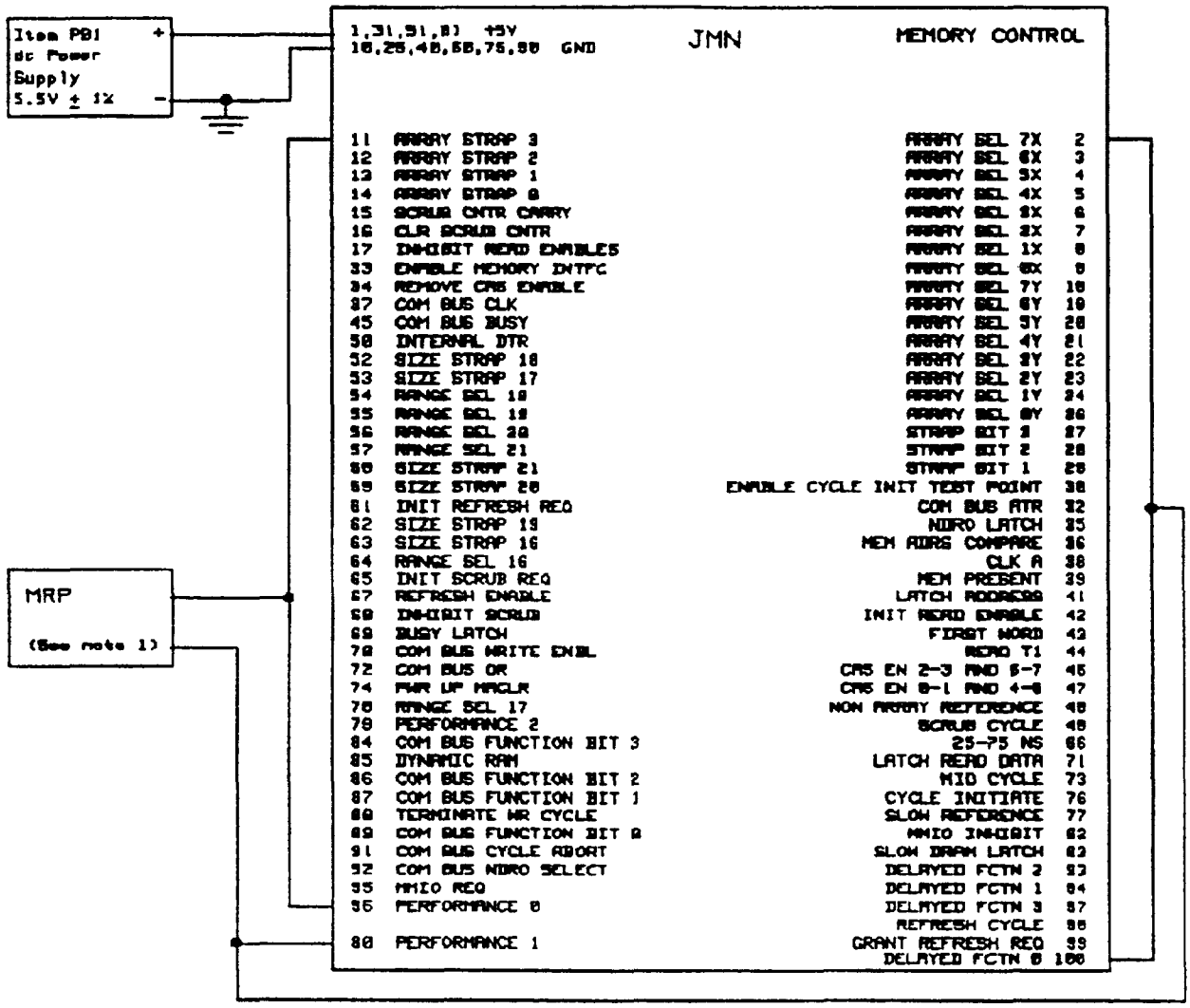
4.5.3 Life test. The life test shall be conducted in accordance with MIL-M-28787. The module shall be operated as shown on figure 5. Following the life test and while still at the maximum class temperature, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the operating temperature end-of-life requirements in table I. Following the return of the module to 25 ± 5°C, electrical tests B, F through K and M through Q shall be performed in the order specified in 4.4.2 using the 25 ± 5°C ambient end-of-life requirements in table I.

4.5.4 Thermal shock test. The thermal shock test shall be conducted in accordance with MIL-M-28787. The module shall be unpowered during this test. Following the thermal shock test, the electrical tests in table I which are marked with an "X" in the test type column, shall be performed using the initial limits at 25 ± 5°C ambient.

5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-M-28787.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 5. JMN life test circuit.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Definitions. Negative (-) current is current directed out of the module pin and positive (+) current is current directed into the module pin. Abbreviations used in this specification are defined as follows:

(a) Cin = input capacitance.
(b) Ci/o = input/output capacitance.
(c) Cout = output capacitance.
(d) Ibd = input breakdown current.
(e) Icc = power supply current.
(f) Iih = input high level current.
(g) Iil = input low level current.
(h) Iin = input forcing current.
(i) Ioh = output high level current.
(j) Iol = output low level current.
(k) Ios = output short circuit current.
(l) Iozh = output high level leakage current.
(m) Iozl = output low level leakage current.
(n) NA = not applicable.
(o) PSFC = power supply filter capacitance.
(p) PSTA = power supply transient amplitude.
(q) Vic = input clamp diode voltage.
(r) Vih = input high level voltage.
(s) Vil = input low level voltage.
(t) Vin = input forcing voltage.
(u) Voh = output high level voltage.
(v) Vol = output low level voltage.
(w) Vout = output forcing voltage.

6.2 End-of-life. When any individual module circuit parameter fails to conform to the end-of-life requirements in table I, the module is considered to have completed its useful life (see 3.9).

6.3 Module description. The following electrical parameters reflect the -55°C to 85°C end-of-life parameters in table I. Operating power supply voltage range: 4.5 V dc to 5.5 V dc. Module power dissipation: 8.58 W (max); 5.75 W (typical).

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6.3.1 Input parameters.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
ARRAYSTRAP 3	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ARRAYSTRAP 2	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ARRAYSTRAP 1	-1.2	0.4	2.4	-0.6	0.02	0.1	75
ARRAYSTRAP 0	-1.2	0.4	2.4	-0.6	0.02	0.1	75
SCRUB CTR CA	-1.2	0.4	2.4	-1.2	0.04	0.2	100
CL SCRUB CTR	-1.2	0.4	2.4	-0.6	0.02	0.1	75
INHIBT RD EN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
EN MEM INTFC	-1.2	0.4	2.4	-0.6	0.02	0.1	75
RMOVE CAS EN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS CLK	-1.2	0.4	2.4	-3.6	0.12	0.6	100
COM BUS BUSY	-1.2	0.4	2.4	-0.6	0.02	0.1	50
INTERNAL DTR	-1.2	0.4	2.4	-2.2	0.07	1.1	75
SIZESTRAP 18	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SIZESTRAP 17	-1.2	0.4	2.4	-0.6	0.02	0.1	50

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
RANGE SEL 18	-1.2	0.4	2.4	-0.6	0.02	0.1	50
RANGE SEL 19	-1.2	0.4	2.4	-0.6	0.02	0.1	50
RANGE SEL 20	-1.2	0.4	2.4	-0.6	0.02	0.1	50
RANGE SEL 21	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SIZESTRAP 21	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SIZESTRAP 20	-1.2	0.4	2.4	-0.6	0.02	0.1	50
IREFRESH REQ	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SIZESTRAP 19	-1.2	0.4	2.4	-0.6	0.02	0.1	50
SIZESTRAP 16	-1.2	0.4	2.4	-0.6	0.02	0.1	50
RANGE SEL 16	-1.2	0.4	2.4	-0.6	0.02	0.1	50
ISCRUB REQ	-1.2	0.4	2.4	-0.6	0.02	0.1	50
REFRESH EN	NA	0.4	2.4	-6.03	-2.85	0.22	75
INHIBITSCRUB	-1.2	0.4	2.4	-0.6	0.02	0.1	50
BUSY LATCH	-1.2	0.4	2.4	-10.6	0.18	1.4	200

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
CBUS WR EN	-1.2	0.4	2.4	-0.6	0.02	0.1	75
COM BUS OR	-1.2	0.4	2.4	-1.8	0.06	0.3	100
PWR UP MACLR	-1.2	0.4	2.4	-0.6	0.02	0.1	75
RANGE SEL 17	-1.2	0.4	2.4	-0.6	0.02	0.1	75
PERFORMANCE2	NA	0.4	2.4	-29.7	-14.3	1.67	200
CBUS FCTN B3	-1.2	0.4	2.4	-1.2	0.04	0.2	100
DYNAMIC RAM	NA	0.4	2.4	-28.3	-14.3	0.78	75
CBUS FCTN B2	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CBUS FCTN B1	-1.2	0.4	2.4	-0.6	0.02	0.1	50
TERMI WR CYC	-1.2	0.4	2.4	-0.6	0.02	0.1	216
CBUS FCTN B0	-1.2	0.4	2.4	-1.2	0.04	0.2	100
CBUS CYC ABT	-1.2	0.4	2.4	-0.6	0.02	0.1	50
CBUS NDRO SL	NA	0.4	2.4	-6.03	-2.85	0.22	75

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6.3.1 Input parameters - Continued.

Signal	Vic min (V)	Vil max (V)	Vih min (V)	Iil min (mA)	Iih max (mA)	Ibd max (mA)	Cin max (pF)
MMIO REQ	NA	0.4	2.4	-27.7	-14.3	0.68	75
PERFORMANCE0	NA	0.4	2.4	-29.7	-14.3	1.67	175
PERFORMANCE1	NA	0.4	2.4	-29.7	-14.3	NA	100

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6.3.2 Output parameters.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
ARRAY SEL 7X	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 6X	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 5X	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 4X	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 3X	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 2X	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 1X	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 0X	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 7Y	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 6Y	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 5Y	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 4Y	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 3Y	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 2Y	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
ARRAY SEL 1Y	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ARRAY SEL 0Y	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
STRAP BIT 3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
STRAP BIT 2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
STRAP BIT 1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
ENCYC TESTPT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
COM BUS ATR	2.4	0.5	-0.4	6.0	0.05	0.05	-100.0	NA
NDRO LATCH	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
MEM AD CPARE	2.4	0.5	-0.4	6.0	0.25	NA	NA	NA
CLK A	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
MEM PRESENT	2.4	0.5	-0.4	6.0	0.5	NA	NA	NA
LATCH ADRESS	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
INIT RD EN	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
FIRST WORD	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
READ T1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CSEN 2-3&6-7	2.4	0.5	-0.4	6.0	NA	NA	-200.0	NA
CSEN 0-1&4-5	2.4	0.5	-0.4	6.0	NA	NA	-200.0	NA
NONARRAY REF	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
SCRUB CYCLE	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
25-75 NS	2.4	0.5	-0.4	6.0	NA	NA	NA	NA
LATCH RDDATA	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
MID CYCLE	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
CYC INITIATE	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
SLOW REF	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
MMIO INHIBIT	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
SLOW DRAM LA	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
DELAYEDFCTN2	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
DELAYEDFCTN1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA

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6.3.2 Output parameters - Continued.

Signal	Voh min (V)	Vol max (V)	Ioh max (mA)	Iol max (mA)	Iozh max (mA)	Iozl max (mA)	Ios max (mA)	Cout max (pF)
DELAYEDFCTN3	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
REFRESH CYC	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
GREFRESH REQ	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
DELAYEDFCTN0	2.4	0.5	-0.4	6.0	NA	NA	-60.0	NA
PERFORMANCE1	2.4	0.5	-0.4	6.0	NA	NA	-60.0	100

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6.3.3 Class I and II dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	COM BUS CLK to SCRUB CYCLE	-	-	-	65	65	65	nS	11
2	tpd	COM BUS CLK to LATCH ADRESS	-	-	-	55	55	55	nS	11
3	tpd	INHIBT RD EN to NONARRAY REF	-	-	-	25	25	25	nS	11
4	tpd	COM BUS CLK to CLK A	-	-	-	14	14	14	nS	11
5	tpd	EN MEM INTFC to MEM PRESENT	-	-	-	26	26	26	nS	11
6	tpd	COM BUS CLK to CSEN 2-3&6-7	-	-	-	47	47	47	nS	11
7	tpd	PERFORMANCE 2-0 to SLOW REF	-	-	-	14	14	14	nS	11
8	tpd	COM BUS CLK to DELAYEDFCTN 3-0	-	-	-	44	44	44	nS	11
8	tpd	COM BUS CLK to SLOW DRAM LA	-	-	-	44	44	44	nS	11
9	tpd	COM BUS CLK to READ T1	-	-	-	61	61	61	nS	11
10	tpd	COM BUS CLK to REFRESH CYC	-	-	-	54	54	54	nS	11

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6.4 Module function. The memory control module contains a range comparator cycle control logic, memory control logic and an array select latch. The range comparator tests incoming memory addresses against hardwired range select bits and strap bits to ensure addresses fall within the range allocated by the user. If the address falls within the proper range, the memory cycle is allowed to start at the receipt of the common bus Originator Ready (OR) under the control of the cycle control logic. The cycle control logic together with the memory control logic generate the internal control timing necessary to carry out the memory operation specified by the common bus function bits. Additionally, the memory control logic tests memory performance bits to ensure the internal control timing initiated is matched to the speed of the memory array modules installed, generates a Column Address Strobe (CAS) enable for the memory array modules and a cycle initiate corresponding to the selected internal control timing. The array select latch output 16 array select (Row Address Strobe (RAS)) signals (1 array select for each of the 16 64K Dynamic Random Access Memory (DRAM) arrays) to the memory array modules.

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6.4.3 Description of I/O pins.

Signal name	Pin number	I/O	Description
ARRAYSTRAP 3	11	I	ARRAY STRAP 3
ARRAYSTRAP 2	12	I	ARRAY STRAP 2
ARRAYSTRAP 1	13	I	ARRAY STRAP 1
ARRAYSTRAP 0	14	I	ARRAY STRAP 0
SCRUB CTR CA	15	I	SCRUB CNTR CARRY
CL SCRUB CTR	16	I	CLR SCRUB CNTR
INHIBT RD EN	17	I	INHIBIT READ ENABLES
EN MEM INTFC	33	I	ENABLE MEMORY INTFC
RMOVE CAS EN	34	I	REMOVE CAS ENABLE
COM BUS CLK	37	I	COM BUS CLK
COM BUS BUSY	45	I	COM BUS BUSY
INTERNAL DTR	50	I	INTERNAL DTR
SIZESTRAP 18	52	I	SIZE STRAP 18
SIZESTRAP 17	53	I	SIZE STRAP 17
RANGE SEL 18	54	I	RANGE SEL 18
RANGE SEL 19	55	I	RANGE SEL 19
RANGE SEL 20	56	I	RANGE SEL 20
RANGE SEL 21	57	I	RANGE SEL 21
SIZESTRAP 21	58	I	SIZE STRAP 21
SIZESTRAP 20	59	I	SIZE STRAP 20

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6.4.3 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
IREFRESH REQ	61	I	INIT REFRESH REQ
SIZESTRAP 19	62	I	SIZE STRAP 19
SIZESTRAP 16	63	I	SIZE STRAP 16
RANGE SEL 16	64	I	RANGE SEL 16
ISCRUB REQ	65	I	INIT SCRUB REQ
REFRESH EN	67	I	REFRESH ENABLE
INHIBITSCRUB	68	I	INHIBIT SCRUB
BUSY LATCH	69	I	BUSY LATCH
CBUS WR EN	70	I	COM BUS WRITE ENABLE
COM BUS OR	72	I	COM BUS OR
PWR UP MACLR	74	I	PWR UP MACLR
RANGE SEL 17	78	I	RANGE SEL 17
PERFORMANCE2	79	I	PERFORMANCE 2
CBUS FCTN B3	84	I	COM BUS FUNCTION BIT 3
DYNAMIC RAM	85	I	DYNAMIC RAM
CBUS FCTN B2	86	I	COM BUS FUNCTION BIT 2
CBUS FCTN B1	87	I	COM BUS FUNCTION BIT 1
TERMI WR CYC	88	I	TERMINATE WR CYCLE
CBUS FCTN B0	89	I	COM BUS FUNCTION BIT 0
CBUS CYC ABT	91	I	COM BUS CYCLE ABORT

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6.4.3 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CBUS NDRO SL	92	I	COM BUS NDRO SELECT
MMIO REQ	95	I	MMIO REQ
PERFORMANCE0	96	I	PERFORMANCE 0
PERFORMANCE1	80	I/O	PERFORMANCE 1
ARRAY SEL 7X	2	O	ARRAY SEL 7X
ARRAY SEL 6X	3	O	ARRAY SEL 6X
ARRAY SEL 5X	4	O	ARRAY SEL 5X
ARRAY SEL 4X	5	O	ARRAY SEL 4X
ARRAY SEL 3X	6	O	ARRAY SEL 3X
ARRAY SEL 2X	7	O	ARRAY SEL 2X
ARRAY SEL 1X	8	O	ARRAY SEL 1X
ARRAY SEL 0X	9	O	ARRAY SEL 0X
ARRAY SEL 7Y	18	O	ARRAY SEL 7Y
ARRAY SEL 6Y	19	O	ARRAY SEL 6Y
ARRAY SEL 5Y	20	O	ARRAY SEL 5Y
ARRAY SEL 4Y	21	O	ARRAY SEL 4Y
ARRAY SEL 3Y	22	O	ARRAY SEL 3Y
ARRAY SEL 2Y	23	O	ARRAY SEL 2Y
ARRAY SEL 1Y	24	O	ARRAY SEL 1Y
ARRAY SEL 0Y	26	O	ARRAY SEL 0Y

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6.4.3 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
STRAP BIT 3	27	O	STRAP BIT 3
STRAP BIT 2	28	O	STRAP BIT 2
STRAP BIT 1	29	O	STRAP BIT 1
ENCYC TESTPT	30	O	ENABLE CYCLE INIT TEST POINT
COM BUS ATR	32	O	COM BUS ATR
NDRO LATCH	35	O	NDRO LATCH
MEM AD CPARE	36	O	MEM ADRS COMPARE
CLK A	38	O	CLK A
MEM PRESENT	39	O	MEM PRESENT
LATCH ADRESS	41	O	LATCH ADDRESS
INIT RD EN	42	O	INIT READ ENABLE
FIRST WORD	43	O	FIRST WORD
READ T1	44	O	READ T1
CSEN 2-3&6-7	46	O	CAS EN 2-3 AND 6-7
CSEN 0-1&4-5	47	O	CAS EN 0-1 AND 4-5
NONARRAY REF	48	O	NON ARRAY REFERENCE
SCRUB CYCLE	49	O	SCRUB CYCLE
25-75 NS	66	O	25-75 NS
LATCH RDDATA	71	O	LATCH READ DATA
MID CYCLE	73	O	MID CYCLE

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6.4.3 Description of I/O pins - Continued.

Signal name	Pin number	I/O	Description
CYC INITIATE	76	O	CYCLE INITIATE
SLOW REF	77	O	SLOW REFERENCE
MMIO INHIBIT	82	O	MMIO INHIBIT
SLOW DRAM LA	83	O	SLOW DRAM LATCH
DELAYEDFCTN2	93	O	DELAYED FCTN 2
DELAYEDFCTN1	94	O	DELAYED FCTN 1
DELAYEDFCTN3	97	O	DELAYED FCTN 3
REFRESH CYC	98	O	REFRESH CYCLE
GREFRESH REQ	99	O	GRANT REFRESH REQ
DELAYEDFCTN0	100	O	DELAYED FCTN 0

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6.5 Functional test vectors. Functional test vectors are not printed herein because of their extreme length; and, even when presented as a hard copy, they require conversion to on-line format. When the entire sequence of vector groups is needed for driving devices as part of inspection, a test program output should be generated in a form compatible with test system architecture (magnetic tape format usually preferred). For reference purposes, the Naval Weapons Support Center (Code 607) maintains a limited quantity of dated, hard copy printouts of the functional test vectors and these may be obtained upon request.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - SH

(Project 5963-0078)

Agent: NW

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TABLE II. Isolation test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
A	Isolation, frame	C	Frame	10	INF	MOhm	1-8

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NOTES:

1. Isolation test tolerance table.

Condition	Value	Accuracy	Test condition
Megohm bridge voltage	100 V	± 10 percent	A
Isolation measurement	NA	± 500 KOhm	A

2. The ohmmeter used shall be item R06.

3. In the set-up conditions in table II, 'A' shall correspond to the positive terminal of the megohm bridge, 'B' shall correspond to the negative terminal and 'C' shall correspond to all used pins shorted together.

4. Warning: When using a megohm bridge, high voltage may be present at any of the output binding posts. Always set the megohm bridge function switch to "DISCHARGE" before connecting or disconnecting the test leads.

5. When testing for frame isolation, penetrate the anodized finish on the module frame with a sharply pointed probe connected to the negative terminal of the megohm bridge.

6. Caution: Set megohm bridge to limit current to 10 uA before applying voltage to the module.

7. The test circuit shall be as indicated on figure 6.

8. Abbreviations used are defined in 6.1.

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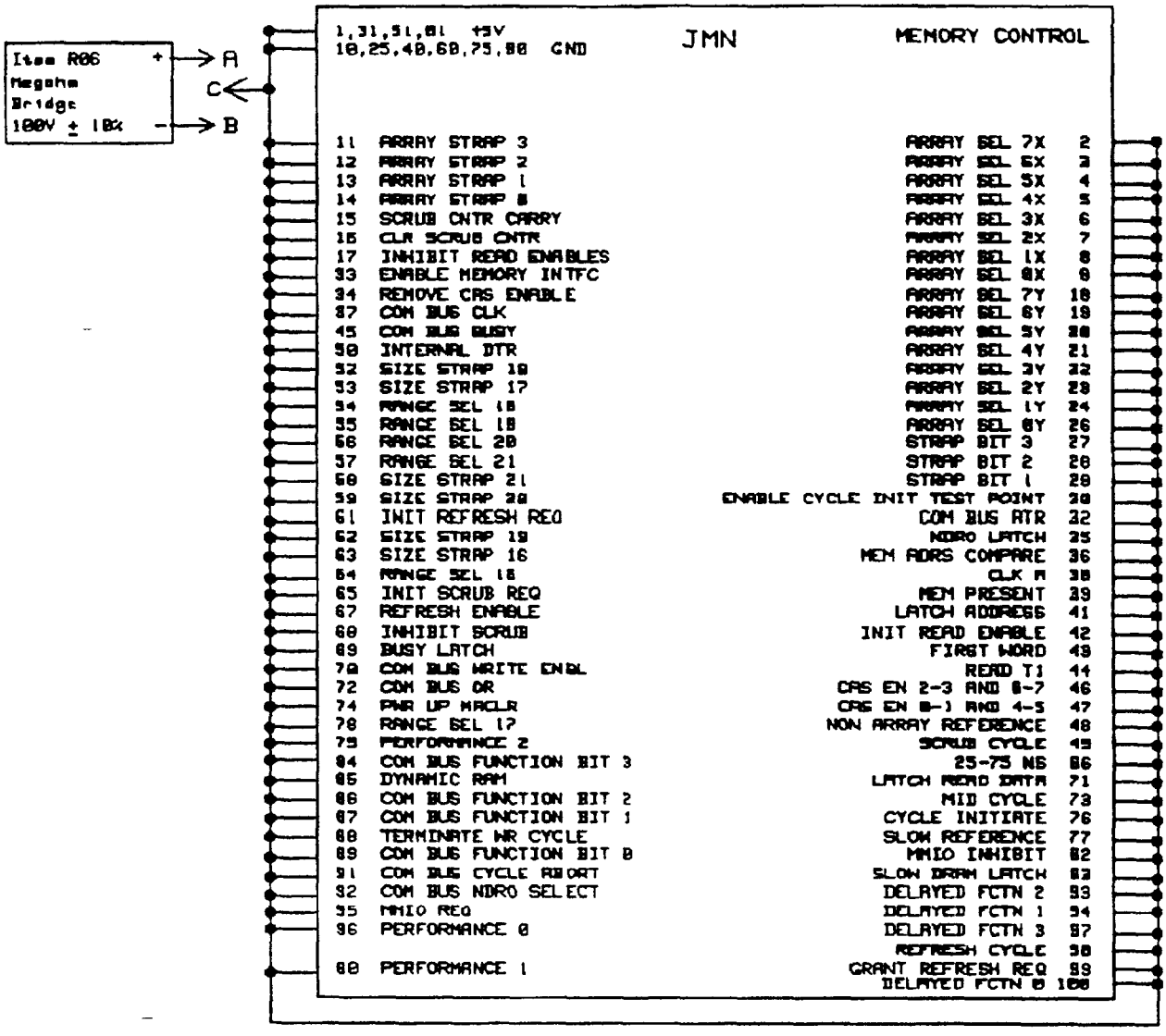


FIGURE 6. Isolation test circuit for table II.

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TABLE III. Continuity test.

Test cond	Test description	Connect A to pin	Connect B to pin	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
B1	Continuity, GND	10	25	0	1	Ohm	1-5
B2	Continuity, GND	10	40				
B3	Continuity, GND	60	75				
B4	Continuity, GND	60	90				
B5	Continuity, Vcc	1	31				
B6	Continuity, Vcc	51	81	0	1	Ohm	1-5

NOTES:

1. Continuity test tolerance table.

Condition	Value	Accuracy	Test condition
Continuity measurement	NA	± 0.05 Ohm	B1-B6

2. The ohmmeter used shall be item R07.

3. In the set-up conditions in table III, 'A' shall correspond to the positive terminal of the ohmmeter, and 'B' shall correspond to the negative terminal.

4. The test circuit shall be as indicated on figure 7.

5. Abbreviations used are defined in 6.1.

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1,31,61,81 +5V 10,25,40,60,75,90 GND		JMN	MEMORY CONTROL
11	ARRAY STRAP 3		ARRAY SEL 7X 2
12	ARRAY STRAP 2		ARRAY SEL 6X 3
13	ARRAY STRAP 1		ARRAY SEL 5X 4
14	ARRAY STRAP 0		ARRAY SEL 4X 5
15	SCRUB CNTR CARRY		ARRAY SEL 3X 6
16	CLR SCRUB CNTR		ARRAY SEL 2X 7
17	INHIBIT READ ENABLES		ARRAY SEL 1X 8
33	ENABLE MEMORY INTFC		ARRAY SEL 0X 9
34	REMOVE CAS ENABLE		ARRAY SEL 7Y 10
37	COM BUS CLK		ARRAY SEL 6Y 19
45	COM BUS BUSY		ARRAY SEL 5Y 20
50	INTERNAL DTR		ARRAY SEL 4Y 21
52	SIZE STRAP 18		ARRAY SEL 3Y 22
53	SIZE STRAP 17		ARRAY SEL 2Y 23
54	RANGE SEL 18		ARRAY SEL 1Y 24
55	RANGE SEL 19		ARRAY SEL 0Y 26
56	RANGE SEL 20		STRAP BIT 3 27
57	RANGE SEL 21		STRAP BIT 2 28
58	SIZE STRAP 21		STRAP BIT 1 29
59	SIZE STRAP 20	ENABLE CYCLE INIT TEST POINT	30
61	INIT REFRESH REQ		COM BUS ATR 32
62	SIZE STRAP 19		NDRO LATCH 35
63	SIZE STRAP 16		MEM ADRS COMPARE 36
64	RANGE SEL 16		CLK A 38
65	INIT SCRUB REQ		MEM PRESENT 39
67	REFRESH ENABLE		LATCH ADDRESS 41
68	INHIBIT SCRUB		INIT READ ENABLE 42
69	BUSY LATCH		FIRST WORD 43
70	COM BUS WRITE ENBL		READ T1 44
72	COM BUS OR		CAS EN 2-3 AND 6-7 46
74	PWR UP MACLR		CAS EN 0-1 AND 4-5 47
78	RANGE SEL 17		NON ARRAY REFERENCE 48
79	PERFORMANCE 2		SCRUB CYCLE 49
84	COM BUS FUNCTION BIT 3		25-75 NS 66
85	DYNAMIC RAM		LATCH READ DATA 71
86	COM BUS FUNCTION BIT 2		MID CYCLE 73
87	COM BUS FUNCTION BIT 1		CYCLE INITIATE 75
88	TERMINATE WR CYCLE		SLOW REFERENCE 77
89	COM BUS FUNCTION BIT 0		MMIO INHIBIT 82
91	COM BUS CYCLE ABORT		SLOW DRAM LATCH 83
92	COM BUS NDRO SELECT		DELAYED FCTN 2 93
95	MMIO REQ		DELAYED FCTN 1 94
96	PERFORMANCE 0		DELAYED FCTN 3 97
			REFRESH CYCLE 98
98	PERFORMANCE 1		GRANT REFRESH REQ 99
			DELAYED FCTN 0 100

FIGURE 7. Continuity test circuit for table III.

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TABLE IV. Power supply stress test.

Test cond	Power supply	Vin (V)	Connect		25 ± 5°C Initial limits	Notes
			Positive lead to	Negative lead to		
C	Vcc	7.0	1,31,51, 81	10,25, 40,60, 75,90	Module shall pass all remaining tests.	1-6

NOTES:

1. Power supply stress test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	7.0 V	+0 mV,-70 mV	C
Power supply current limit	2.808 A	NA	C

2. There shall be no power supply voltage overshoot on this test.

3. Connect the power supplies to the module as indicated in table IV.

4. When performing the power supply stress test, no measurements are taken. This condition shall be maintained for a minimum of 5 seconds.

5. The test circuit shall be as indicated on figure 8.

6. Abbreviations used are defined in 6.1.

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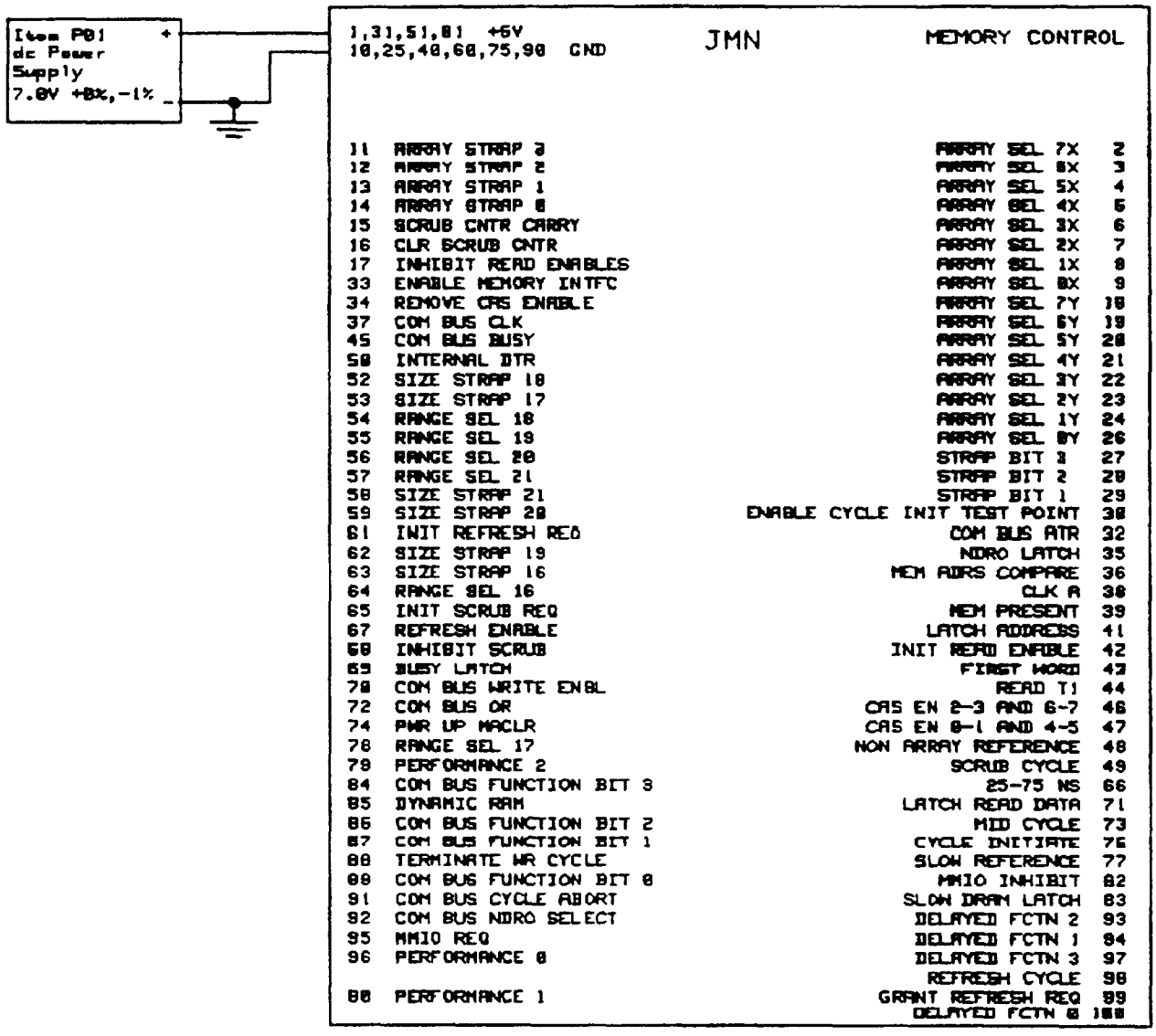


FIGURE 8. Vcc stress test circuit for table IV.

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TABLE V. Input breakdown current test.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D1	ARRAYSTRAP 3	11	5.5	0.0	0.1	mA	1-3
D2	ARRAYSTRAP 2	12					
D3	ARRAYSTRAP 1	13					
D4	ARRAYSTRAP 0	14	5.5	0.0	0.1	mA	1-3
D5	SCRUB CTR CA	15	5.5	0.0	0.2	mA	1-3
D6	CL SCRUB CTR	16	5.5	0.0	0.1	mA	1-3
D7	INHIBT RD EN	17					
D8	EN MEM INTFC	33					
D9	RMOVE CAS EN	34	5.5	0.0	0.1	mA	1-3
D10	COM BUS CLK	37	5.5	0.0	0.6	mA	1-3
D11	COM BUS BUSY	45	5.5	0.0	0.1	mA	1-3
D12	INTERNAL DTR	50	5.5	0.0	1.1	mA	1-3
D13	SIZESTRAP 18	52	5.5	0.0	0.1	mA	1-3
D14	SIZESTRAP 17	53					
D15	RANGE SEL 18	54					
D16	RANGE SEL 19	55					
D17	RANGE SEL 20	56					
D18	RANGE SEL 21	57					
D19	SIZESTRAP 21	58					
D20	SIZESTRAP 20	59	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D21	IREFRESH REQ	61	5.5	0.0	0.1	mA	1-3
D22	SIZESTRAP 19	62					
D23	SIZESTRAP 16	63					
D24	RANGE SEL 16	64					
D25	ISCRUB REQ	65	5.5	0.0	0.1	mA	1-3
D26	REFRESH EN	67	5.5	-0.12	0.22	mA	1-3
D27	INHIBITSCRUB	68	5.5	0.0	0.1	mA	1-3
D28	BUSY LATCH	69	5.5	0.0	1.4	mA	1-3
D29	CBUS WR EN	70	5.5	0.0	0.1	mA	1-3
D30	COM BUS OR	72	5.5	0.0	0.3	mA	1-3
D31	PWR UP MACLR	74	5.5	0.0	0.1	mA	1-3
D32	RANGE SEL 17	78	5.5	0.0	0.1	mA	1-3
D33	PERFORMANCE2	79	5.5	-0.58	1.67	mA	1-3
D34	CBUS FCTN B3	84	5.5	0.0	0.2	mA	1-3
D35	DYNAMIC RAM	85	5.5	-0.58	0.78	mA	1-3
D36	CBUS FCTN B2	86	5.5	0.0	0.1	mA	1-3
D37	CBUS FCTN B1	87					
D38	TERMI WR CYC	88	5.5	0.0	0.1	mA	1-3
D39	CBUS FCTN B0	89	5.5	0.0	0.2	mA	1-3
D40	CBUS CYC ABT	91	5.5	0.0	0.1	mA	1-3

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TABLE V. Input breakdown current test - Continued.

Test cond	Signal name	Pin no.	Vin (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
D41	CBUS NDRO SL	92	5.5	-0.12	0.22	mA	1-3
D42	MMIO REQ	95	5.5	-0.58	0.68	mA	1-3
D43	PERFORMANCE0	96	5.5	-0.58	1.67	mA	1-3

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NOTES:

1. Input breakdown current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	D1-D43
Power supply current limit	2.34 A	NA	D1-D43
Forcing voltage, (Vin)	5.5 V	± 55 mV	D1-D43
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 D1-D43 D1-D43
Input current measurement	NA	± 5 uA	D1-D4
	NA	± 10 uA	D5
	NA	± 5 uA	D6-D9
	NA	± 30 uA	D10
	NA	± 5 uA	D11
	NA	± 55 uA	D12
	NA	± 5 uA	D13-D25
	NA	± 17 uA	D26
	NA	± 5 uA	D27
	NA	± 70 uA	D28
	NA	± 5 uA	D29
	NA	± 15 uA	D30
	NA	± 5 uA	D31,D32
	NA	± 112 uA	D33
	NA	± 10 uA	D34
	NA	± 68 uA	D35
	NA	± 5 uA	D36-D38
	NA	± 10 uA	D39
	NA	± 5 uA	D40
	NA	± 17 uA	D41
	NA	± 63 uA	D42
	NA	± 112 uA	D43

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table V and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VI. Output short circuit current test.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E1	ARRAY SEL 7X	2	0.0	-150.0	-60.0	mA	1-4
E2	ARRAY SEL 7X	2	0.0	-350.0	-60.0	mA	1-4
E3	ARRAY SEL 6X	3	0.0	-150.0	-60.0	mA	1-4
E4	ARRAY SEL 6X	3	0.0	-350.0	-60.0	mA	1-4
E5	ARRAY SEL 5X	4	0.0	-150.0	-60.0	mA	1-4
E6	ARRAY SEL 5X	4	0.0	-350.0	-60.0	mA	1-4
E7	ARRAY SEL 4X	5	0.0	-150.0	-60.0	mA	1-4
E8	ARRAY SEL 4X	5	0.0	-350.0	-60.0	mA	1-4
E9	ARRAY SEL 3X	6	0.0	-350.0	-60.0	mA	1-4
E10	ARRAY SEL 3X	6	0.0	-150.0	-60.0	mA	1-4
E11	ARRAY SEL 2X	7	0.0	-350.0	-60.0	mA	1-4
E12	ARRAY SEL 2X	7	0.0	-150.0	-60.0	mA	1-4
E13	ARRAY SEL 1X	8	0.0	-350.0	-60.0	mA	1-4
E14	ARRAY SEL 1X	8	0.0	-150.0	-60.0	mA	1-4
E15	ARRAY SEL 0X	9	0.0	-350.0	-60.0	mA	1-4
E16	ARRAY SEL 0X	9	0.0	-150.0	-60.0	mA	1-4
E17	ARRAY SEL 7Y	18	0.0	-150.0	-60.0	mA	1-4
E18	ARRAY SEL 7Y	18	0.0	-350.0	-60.0	mA	1-4
E19	ARRAY SEL 6Y	19	0.0	-150.0	-60.0	mA	1-4
E20	ARRAY SEL 6Y	19	0.0	-350.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E21	ARRAY SEL 5Y	20	0.0	-150.0	-60.0	mA	1-4
E22	ARRAY SEL 5Y	20	0.0	-350.0	-60.0	mA	1-4
E23	ARRAY SEL 4Y	21	0.0	-150.0	-60.0	mA	1-4
E24	ARRAY SEL 4Y	21	0.0	-350.0	-60.0	mA	1-4
E25	ARRAY SEL 3Y	22	0.0	-350.0	-60.0	mA	1-4
E26	ARRAY SEL 3Y	22	0.0	-150.0	-60.0	mA	1-4
E27	ARRAY SEL 2Y	23	0.0	-350.0	-60.0	mA	1-4
E28	ARRAY SEL 2Y	23	0.0	-150.0	-60.0	mA	1-4
E29	ARRAY SEL 1Y	24	0.0	-350.0	-60.0	mA	1-4
E30	ARRAY SEL 1Y	24	0.0	-150.0	-60.0	mA	1-4
E31	ARRAY SEL 0Y	26	0.0	-350.0	-60.0	mA	1-4
E32	ARRAY SEL 0Y	26	0.0	-150.0	-60.0	mA	1-4
E33	STRAP BIT 3	27					
E34	STRAP BIT 2	28					
E35	STRAP BIT 1	29					
E36	ENCYC TESTPT	30	0.0	-150.0	-60.0	mA	1-4
E37	COM BUS ATR	32	0.0	-325.0	-100.0	mA	1-4
E38	NDRO LATCH	35	0.0	-150.0	-60.0	mA	1-4
E39	CLK A	38					
E40	LATCH ADRESS	41	0.0	-150.0	-60.0	mA	1-4

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TABLE VI. Output short circuit current test - Continued.

Test cond	Signal name	Pin no.	Vout (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
E41	INIT RD EN	42	0.0	-150.0	-60.0	mA	1-4
E42	FIRST WORD	43					
E43	READ T1	44	0.0	-150.0	-60.0	mA	1-4
E44	CSEN 2-3&6-7	46	0.0	-650.0	-200.0	mA	1-4
E45	CSEN 0-1&4-5	47	0.0	-650.0	-200.0	mA	1-4
E46	NONARRAY REF	48	0.0	-150.0	-60.0	mA	1-4
E47	SCRUB CYCLE	49	0.0	-350.0	-60.0	mA	1-4
E48	LATCH RDDATA	71	0.0	-150.0	-60.0	mA	1-4
E49	MID CYCLE	73					
E50	CYC INITIATE	76					
E51	SLOW REF	77					
E52	MMIO INHIBIT	82					
E53	SLOW DRAM LA	83					
E54	DELAYEDFCTN2	93					
E55	DELAYEDFCTN1	94					
E56	DELAYEDFCTN3	97					
E57	REFRESH CYC	98					
E58	GREFRESH REQ	99					
E59	DELAYEDFCTN0	100	0.0	-150.0	-60.0	mA	1-4

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NOTES:

1. Output short circuit current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	E1-E59
Power supply current limit	2.34 A	NA	E1-E59
Forcing voltage, (Vout)	0.0 V	± 50 mV	E1-E59
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	E1-E59
low, (VL)	0.0 V	± 100 mV	E1-E59
Output current measurement	NA	± 4.5 mA	E1
	NA	± 14.5 mA	E2
	NA	± 4.5 mA	E3
	NA	± 14.5 mA	E4
	NA	± 4.5 mA	E5
	NA	± 14.5 mA	E6
	NA	± 4.5 mA	E7
	NA	± 14.5 mA	E8,E9
	NA	± 4.5 mA	E10
	NA	± 14.5 mA	E11
	NA	± 4.5 mA	E12
	NA	± 14.5 mA	E13
	NA	± 4.5 mA	E14
	NA	± 14.5 mA	E15
	NA	± 4.5 mA	E16,E17
	NA	± 14.5 mA	E18
	NA	± 4.5 mA	E19
	NA	± 14.5 mA	E20
	NA	± 4.5 mA	E21

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output current measurement	NA	± 14.5 mA	E22
	NA	± 4.5 mA	E23
	NA	± 14.5 mA	E24,E25
	NA	± 4.5 mA	E26
	NA	± 14.5 mA	E27
	NA	± 4.5 mA	E28
	NA	± 14.5 mA	E29
	NA	± 4.5 mA	E30
	NA	± 14.5 mA	E31
	NA	± 4.5 mA	E32-E36
	NA	± 11.2 mA	E37
	NA	± 4.5 mA	E38-E43
	NA	± 22.5 mA	E44,E45
	NA	± 4.5 mA	E46
	NA	± 14.5 mA	E47
	NA	± 4.5 mA	E48-E59

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VI and the current then measured. The forcing voltage undershoot shall not exceed 100 mV.

3. When performing this test, only one output shall be tested at a time, and the duration of momentarily grounding each output shall not exceed 1 second.

4. Abbreviations used are defined in 6.1.

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TABLE VII. Input current, high level, test.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F1	ARRAYSTRAP 3	11	2.4	-0.6	0.02	mA	1-3
F2	ARRAYSTRAP 2	12					
F3	ARRAYSTRAP 1	13					
F4	ARRAYSTRAP 0	14	2.4	-0.6	0.02	mA	1-3
F5	SCRUB CTR CA	15	2.4	-1.2	0.04	mA	1-3
F6	CL SCRUB CTR	16	2.4	-0.6	0.02	mA	1-3
F7	INHIBT RD EN	17					
F8	EN MEM INTFC	33					
F9	RMOVE CAS EN	34	2.4	-0.6	0.02	mA	1-3
F10	COM BUS CLK	37	2.4	-3.6	0.12	mA	1-3
F11	COM BUS BUSY	45	2.4	-0.6	0.02	mA	1-3
F12	INTERNAL DTR	50	2.4	-2.2	0.07	mA	1-3
F13	SIZESTRAP 18	52	2.4	-0.6	0.02	mA	1-3
F14	SIZESTRAP 17	53					
F15	RANGE SEL 18	54					
F16	RANGE SEL 19	55					
F17	RANGE SEL 20	56					
F18	RANGE SEL 21	57					
F19	SIZESTRAP 21	58					
F20	SIZESTRAP 20	59	2.4	-0.6	0.02	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F21	IREFRESH REQ	61	2.4	-0.6	0.02	mA	1-3
F22	SIZESTRAP 19	62					
F23	SIZESTRAP 16	63					
F24	RANGE SEL 16	64					
F25	ISCRUB REQ	65	2.4	-0.6	0.02	mA	1-3
F26	REFRESH EN	67	2.4	-6.03	-2.85	mA	1-3
F27	INHIBITSCRUB	68	2.4	-0.6	0.02	mA	1-3
F28	BUSY LATCH	69	2.4	-10.6	0.18	mA	1-3
F29	CBUS WR EN	70	2.4	-0.6	0.02	mA	1-3
F30	COM BUS OR	72	2.4	-1.8	0.06	mA	1-3
F31	PWR UP MACLR	74	2.4	-0.6	0.02	mA	1-3
F32	RANGE SEL 17	78	2.4	-0.6	0.02	mA	1-3
F33	PERFORMANCE2	79	2.4	-29.7	-14.3	mA	1-3
F34	PERFORMANCE1	80	2.4	-29.7	-14.3	mA	1-3
F35	CBUS FCTN B3	84	2.4	-1.2	0.04	mA	1-3
F36	DYNAMIC RAM	85	2.4	-28.3	-14.3	mA	1-3
F37	CBUS FCTN B2	86	2.4	-0.6	0.02	mA	1-3
F38	CBUS FCTN B1	87					
F39	TERMI WR CYC	88	2.4	-0.6	0.02	mA	1-3
F40	CBUS FCTN B0	89	2.4	-1.2	0.04	mA	1-3

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TABLE VII. Input current, high level, test - Continued.

Test cond	Signal name	Pin no.	Vih (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
F41	CBUS CYC ABT	91	2.4	-0.6	0.02	mA	1-3
F42	CBUS NDRO SL	92	2.4	-6.03	-2.85	mA	1-3
F43	MMIO REQ	95	2.4	-27.7	-14.3	mA	1-3
F44	PERFORMANCE0	96	2.4	-29.7	-14.3	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:16Z
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NOTES:

1. Input current, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	F1-F44
Power supply current limit	2.34 A	NA	F1-F44
Forcing voltage, (Vih)	2.4 V	± 50 mV	F1-F44
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 F1-F44 F1-F44
Input current measurement	NA	± 31.0 uA	F1-F4
	NA	± 62.0 uA	F5
	NA	± 31.0 uA	F6-F9
	NA	± 186.0 uA	F10
	NA	± 31.0 uA	F11
	NA	± 113.0 uA	F12
	NA	± 31.0 uA	F13-F25
	NA	± 159.0 uA	F26
	NA	± 31.0 uA	F27
	NA	± 540.0 uA	F28
	NA	± 31.0 uA	F29
	NA	± 93.0 uA	F30
	NA	± 31.0 uA	F31,F32
	NA	± 770.0 uA	F33,F34
	NA	± 62.0 uA	F35
	NA	± 700.0 uA	F36
	NA	± 31.0 uA	F37-F39
	NA	± 62.0 uA	F40
	NA	± 31.0 uA	F41
	NA	± 159.0 uA	F42
	NA	± 670.0 uA	F43
	NA	± 770.0 uA	F44

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Check the source to verify that this is the current version before use.

NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE VIII. Input current, low level, test.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G1	ARRAYSTRAP 3	11	0.4	-0.6	0.02	mA	1-3
G2	ARRAYSTRAP 2	12					
G3	ARRAYSTRAP 1	13					
G4	ARRAYSTRAP 0	14	0.4	-0.6	0.02	mA	1-3
G5	SCRUB CTR CA	15	0.4	-1.2	0.04	mA	1-3
G6	CL SCRUB CTR	16	0.4	-0.6	0.02	mA	1-3
G7	INHIBT RD EN	17					
G8	EN MEM INTFC	33					
G9	RMOVE CAS EN	34	0.4	-0.6	0.02	mA	1-3
G10	COM BUS CLK	37	0.4	-3.6	0.12	mA	1-3
G11	COM BUS BUSY	45	0.4	-0.6	0.02	mA	1-3
G12	INTERNAL DTR	50	0.4	-2.2	0.07	mA	1-3
G13	SIZESTRAP 18	52	0.4	-0.6	0.02	mA	1-3
G14	SIZESTRAP 17	53					
G15	RANGE SEL 18	54					
G16	RANGE SEL 19	55					
G17	RANGE SEL 20	56					
G18	RANGE SEL 21	57					
G19	SIZESTRAP 21	58					
G20	SIZESTRAP 20	59	0.4	-0.6	0.02	mA	1-3

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TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G21	IREFRESH REQ	61	0.4	-0.6	0.02	mA	1-3
G22	SIZESTRAP 19	62					
G23	SIZESTRAP 16	63					
G24	RANGE SEL 16	64					
G25	ISCRUB REQ	65	0.4	-0.6	0.02	mA	1-3
G26	REFRESH EN	67	0.4	-6.03	-2.85	mA	1-3
G27	INHIBITSCRUB	68	0.4	-0.6	0.02	mA	1-3
G28	BUSY LATCH	69	0.4	-10.6	0.18	mA	1-3
G29	CBUS WR EN	70	0.4	-0.6	0.02	mA	1-3
G30	COM BUS OR	72	0.4	-1.8	0.06	mA	1-3
G31	PWR UP MACLR	74	0.4	-0.6	0.02	mA	1-3
G32	RANGE SEL 17	78	0.4	-0.6	0.02	mA	1-3
G33	PERFORMANCE2	79	0.4	-29.7	-14.3	mA	1-3
G34	PERFORMANCE1	80	0.4	-29.7	-14.3	mA	1-3
G35	CBUS FCTN B3	84	0.4	-1.2	0.04	mA	1-3
G36	DYNAMIC RAM	85	0.4	-28.3	-14.3	mA	1-3
G37	CBUS FCTN B2	86	0.4	-0.6	0.02	mA	1-3
G38	CBUS FCTN B1	87					
G39	TERMI WR CYC	88	0.4	-0.6	0.02	mA	1-3
G40	CBUS FCTN B0	89	0.4	-1.2	0.04	mA	1-3

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Check the source to verify that this is the current version before use.

TABLE VIII. Input current, low level, test - Continued.

Test cond	Signal name	Pin no.	Vil (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
G41	CBUS CYC ABT	91	0.4	-0.6	0.02	mA	1-3
G42	CBUS NDRO SL	92	0.4	-6.03	-2.85	mA	1-3
G43	MMIO REQ	95	0.4	-27.7	-14.3	mA	1-3
G44	PERFORMANCE0	96	0.4	-29.7	-14.3	mA	1-3

Source: https://assist.dla.mil -- Downloaded: 2026-10-20T18:16Z
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NOTES:

1. Input current, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	G1-G44
Power supply current limit	2.34 A	NA	G1-G44
Forcing voltage, (Vil)	0.4 V	± 50 mV	G1-G44
Pattern voltage			
high, (VH)	3.0 V	± 100 mV	G1-G44
low, (VL)	0.0 V	± 100 mV	G1-G44
Input current measurement	NA	± 31.0 uA	G1-G4
	NA	± 62.0 uA	G5
	NA	± 31.0 uA	G6-G9
	NA	± 186.0 uA	G10
	NA	± 31.0 uA	G11
	NA	± 113.0 uA	G12
	NA	± 31.0 uA	G13-G25
	NA	± 159.0 uA	G26
	NA	± 31.0 uA	G27
	NA	± 540.0 uA	G28
	NA	± 31.0 uA	G29
	NA	± 93.0 uA	G30
	NA	± 31.0 uA	G31,G32
	NA	± 770.0 uA	G33,G34
	NA	± 62.0 uA	G35
	NA	± 700.0 uA	G36
	NA	± 31.0 uA	G37-G39
	NA	± 62.0 uA	G40
	NA	± 31.0 uA	G41
	NA	± 159.0 uA	G42
	NA	± 670.0 uA	G43
	NA	± 770.0 uA	G44

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table VIII and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE IX. Output voltage, high level, test.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H1	ARRAY SEL 7X	2	-0.4	2.4	4.5	V	1-4
H2	ARRAY SEL 7X	2					
H3	ARRAY SEL 6X	3					
H4	ARRAY SEL 6X	3					
H5	ARRAY SEL 5X	4					
H6	ARRAY SEL 5X	4					
H7	ARRAY SEL 4X	5					
H8	ARRAY SEL 4X	5					
H9	ARRAY SEL 3X	6					
H10	ARRAY SEL 3X	6					
H11	ARRAY SEL 2X	7					
H12	ARRAY SEL 2X	7					
H13	ARRAY SEL 1X	8					
H14	ARRAY SEL 1X	8					
H15	ARRAY SEL 0X	9					
H16	ARRAY SEL 0X	9					
H17	ARRAY SEL 7Y	18					
H18	ARRAY SEL 7Y	18					
H19	ARRAY SEL 6Y	19					
H20	ARRAY SEL 6Y	19	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H21	ARRAY SEL 5Y	20	-0.4	2.4	4.5	V	1-4
H22	ARRAY SEL 5Y	20					
H23	ARRAY SEL 4Y	21					
H24	ARRAY SEL 4Y	21					
H25	ARRAY SEL 3Y	22					
H26	ARRAY SEL 3Y	22					
H27	ARRAY SEL 2Y	23					
H28	ARRAY SEL 2Y	23					
H29	ARRAY SEL 1Y	24					
H30	ARRAY SEL 1Y	24					
H31	ARRAY SEL 0Y	26					
H32	ARRAY SEL 0Y	26					
H33	STRAP BIT 3	27					
H34	STRAP BIT 2	28					
H35	STRAP BIT 1	29					
H36	ENCYC TESTPT	30					
H37	COM BUS ATR	32					
H38	NDRO LATCH	35					
H39	CLK A	38					
H40	LATCH ADRESS	41	-0.4	2.4	4.5	V	1-4

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TABLE IX. Output voltage, high level, test - Continued.

Test cond	Signal name	Pin no.	Ioh (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
H41	INIT RD EN	42	-0.4	2.4	4.5	V	1-4
H42	FIRST WORD	43					
H43	READ T1	44					
H44	CSEN 2-3&6-7	46					
H45	CSEN 0-1&4-5	47					
H46	NONARRAY REF	48					
H47	SCRUB CYCLE	49					
H48	25-75 NS	66					
H49	LATCH RDDATA	71					
H50	MID CYCLE	73					
H51	CYC INITIATE	76					
H52	SLOW REF	77					
H53	MMIO INHIBIT	82					
H54	SLOW DRAM LA	83					
H55	DELAYEDFCTN2	93					
H56	DELAYEDFCTN1	94					
H57	DELAYEDFCTN3	97					
H58	REFRESH CYC	98					
H59	GREFRESH REQ	99					
H60	DELAYEDFCTN0	100	-0.4	2.4	4.5	V	1-4

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NOTES:

1. Output voltage, high level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	H1-H60
Power supply current limit	2.34 A	NA	H1-H60
Forcing current, (Ioh)	-0.4 mA	± 40 uA	H1-H60
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 H1-H60 H1-H60
Output voltage measurement	NA	± 105 mV	H1-H60

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

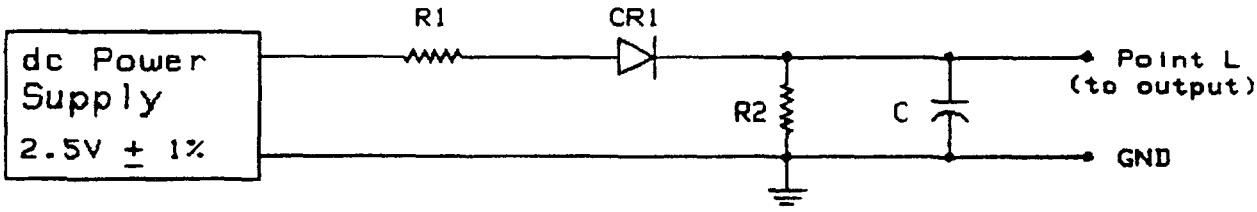
(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table IX and the voltage then measured.

3. The load shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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NOTES:

1. CR1 = 1N4148 or equivalent.

2. The load capacitance is defined as follows:

(a) C = 100 ± 10 pF (including probe and parasitic capacitance) for sample dynamic tests.

(b) C = 300 ± 30 pF (including probe and parasitic capacitance) for qualification dynamic tests.

(c) The functional test shall be performed with only a 100 ± 10 pF load (including probe and parasitic capacitance) attached to each output.

(d) C not required for any other tests.

FIGURE 9. Logic load circuit.

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NOTES: - Continued.

3. Select R1 for a current flow of Iol when point L is held at Vol. The values of Iol and Vol for each output are in the table below.

4. The value of R2 for each output load shall be as specified in the table below.

Pin number	R2 1/ (ohms)	Iol (mA)	Vol (volts)
2,3,4,5,6,7,8,9, 18,19,20,21,22, 23,24,26,27,28, 29,30,32,35,38, 41,42,43,44,46, 47,48,49,66,71, 73,76,77,82,83, 93,94,97,98,99, 100	6200	6.0	0.5
36,39,80	NA	6.0	0.5

1/ All resistors are metal film, ± 1 percent, 1/4 W.

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TABLE X. Output voltage, low level, test.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	25 ± 5°C Initial limits Max	Units	Notes
J1	ARRAY SEL 7X	2	6.0	0.0	0.5	V	1-4
J2	ARRAY SEL 7X	2					
J3	ARRAY SEL 6X	3					
J4	ARRAY SEL 6X	3					
J5	ARRAY SEL 5X	4					
J6	ARRAY SEL 5X	4					
J7	ARRAY SEL 4X	5					
J8	ARRAY SEL 4X	5					
J9	ARRAY SEL 3X	6					
J10	ARRAY SEL 3X	6					
J11	ARRAY SEL 2X	7					
J12	ARRAY SEL 2X	7					
J13	ARRAY SEL 1X	8					
J14	ARRAY SEL 1X	8					
J15	ARRAY SEL 0X	9					
J16	ARRAY SEL 0X	9					
J17	ARRAY SEL 7Y	18					
J18	ARRAY SEL 7Y	18					
J19	ARRAY SEL 6Y	19					
J20	ARRAY SEL 6Y	19	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J21	ARRAY SEL 5Y	20	6.0	0.0	0.5	V	1-4
J22	ARRAY SEL 5Y	20					
J23	ARRAY SEL 4Y	21					
J24	ARRAY SEL 4Y	21					
J25	ARRAY SEL 3Y	22					
J26	ARRAY SEL 3Y	22					
J27	ARRAY SEL 2Y	23					
J28	ARRAY SEL 2Y	23					
J29	ARRAY SEL 1Y	24					
J30	ARRAY SEL 1Y	24					
J31	ARRAY SEL 0Y	26					
J32	ARRAY SEL 0Y	26					
J33	STRAP BIT 3	27					
J34	STRAP BIT 2	28					
J35	STRAP BIT 1	29					
J36	ENCYC TESTPT	30					
J37	COM BUS ATR	32					
J38	NDRO LATCH	35					
J39	MEM AD CPARE	36					
J40	CLK A	38	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J41	MEM PRESENT	39	6.0	0.0	0.5	V	1-4
J42	LATCH ADRESS	41					
J43	INIT RD EN	42					
J44	FIRST WORD	43					
J45	READ T1	44					
J46	CSEN 2-3&6-7	46					
J47	CSEN 0-1&4-5	47					
J48	NONARRAY REF	48					
J49	SCRUB CYCLE	49					
J50	25-75 NS	66					
J51	LATCH RDDATA	71					
J52	MID CYCLE	73					
J53	CYC INITIATE	76					
J54	SLOW REF	77					
J55	PERFORMANCE1	80					
J56	MMIO INHIBIT	82					
J57	SLOW DRAM LA	83					
J58	DELAYEDFCTN2	93					
J59	DELAYEDFCTN1	94					
J60	DELAYEDFCTN3	97	6.0	0.0	0.5	V	1-4

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TABLE X. Output voltage, low level, test - Continued.

Test cond	Signal name	Pin no.	Iol (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
J61	REFRESH CYC	98	6.0	0.0	0.5	V	1-4
J62	GREFRESH REQ	99	″	″	″	″	″
J63	DELAYEDFCTN0	100	6.0	0.0	0.5	V	1-4

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NOTES:

1. Output voltage, low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V	± 45 mV	J1-J63
Power supply current limit	2.34 A	NA	J1-J63
Forcing current, (Iol)	6.0 mA	± 600 uA	J1-J63
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 J1-J63 J1-J63
Output voltage measurement	NA	± 25 mV	J1-J63

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing current shall be applied to the pin specified in the appropriate test conditions in table X and the voltage then measured.

3. The lead shall be as indicated on figure 9.

4. Abbreviations used are defined in 6.1.

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TABLE XI. Output leakage current, high and low level, test.

Test cond	Signal name	Pin no.	Forcing voltage (V)	25 ± 5°C Initial limits Min	Max	Units	Notes
High level							
K1	COM BUS ATR	32	2.4	-0.05	0.05	mA	1-3
K2	MEM AD CPARE	36	5.5	0.0	0.25	mA	1-3
K3	MEM PRESENT	39	5.5	0.0	0.5	mA	1-3
Low level							
K4	COM BUS ATR	32	0.4	-0.05	0.05	mA	1-3

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NOTES:

1. Output leakage current, high and low level, test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	K1-K4
Power supply current limit	2.34 A	NA	K1-K4
Forcing voltage, (Voh)	2.4 V	± 50 mV	K1
(Voh)	5.5 V	± 50 mV	K2,K3
(Vol)	0.4 V	± 50 mV	K4
Pattern voltage high, (VH)	3.0 V	± 100 mV	K1-K4
low, (VL)	0.0 V	± 100 mV	K1-K4
Output current measurement	NA	± 5 uA	K1
	NA	± 12.5 uA	K2
	NA	± 25 uA	K3
	NA	± 5 uA	K4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test to achieve the required state and make the indicated measurement at the appropriate pattern line.

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NOTES: - Continued.

(c) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(d) The forcing voltage shall be applied to the pin specified in the appropriate test conditions in table XI and the current then measured. The forcing voltage overshoot shall not exceed 100 mV.

3. Abbreviations used are defined in 6.1.

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TABLE XII. Input clamp diode voltage test.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L1	ARRAYSTRAP 3	11	-18.0	-1.2	0.0	V	1-3
L2	ARRAYSTRAP 2	12					
L3	ARRAYSTRAP 1	13					
L4	ARRAYSTRAP 0	14					
L5	SCRUB CTR CA	15					
L6	CL SCRUB CTR	16					
L7	INHIBT RD EN	17					
L8	EN MEM INTFC	33					
L9	RMOVE CAS EN	34					
L10	COM BUS CLK	37					
L11	COM BUS BUSY	45					
L12	INTERNAL DTR	50					
L13	SIZESTRAP 18	52					
L14	SIZESTRAP 17	53					
L15	RANGE SEL 18	54					
L16	RANGE SEL 19	55					
L17	RANGE SEL 20	56					
L18	RANGE SEL 21	57					
L19	SIZESTRAP 21	58					
L20	SIZESTRAP 20	59	-18.0	-1.2	0.0	V	1-3

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TABLE XII. Input clamp diode voltage test - Continued.

Test cond	Signal name	Pin no.	Iin (mA)	25 ± 5°C Initial limits Min	Max	Units	Notes
L21	IREFRESH REQ	61	-18.0	-1.2	0.0	V	1-3
L22	SIZESTRAP 19	62					
L23	SIZESTRAP 16	63					
L24	RANGE SEL 16	64					
L25	ISCRUB REQ	65					
L26	INHIBITSCRUB	68					
L27	BUSY LATCH	69					
L28	CBUS WR EN	70					
L29	COM BUS OR	72					
L30	PWR UP MACLR	74					
L31	RANGE SEL 17	78					
L32	CBUS FCTN B3	84					
L33	CBUS FCTN B2	86					
L34	CBUS FCTN B1	87					
L35	TERMI WR CYC	88					
L36	CBUS FCTN B0	89					
L37	CBUS CYC ABT	91	-18.0	-1.2	0.0	V	1-3

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NOTES:

1. Input clamp diode voltage test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	Off	NA	L1-L37
Power supply current limit	NA	NA	L1-L37
Forcing current, (Iin)	-18 mA	± 1.8 mA	L1-L37
Pattern voltage high, (VH) low, (VL)	 NA NA	 NA NA	 L1-L37 L1-L37
Input voltage measurement	NA	± 60 mV	L1-L37

2. The following procedure is required to perform this test:

(a) No power shall be applied to the module.

(b) When making the test indicated on a pin, insure that the pin is disconnected from any drivers or comparators except those called for in the test.

(c) The forcing current shall be applied to the pin specified in the appropriate test conditions in table XII and the voltage then measured.

3. Abbreviations used are defined in 6.1.

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TABLE XIII. Power supply filter capacitance test.

Test cond	Parameter	Pin numbers	25 ± 5°C Initial limits Min	Max	Units	Notes
M1	PSFC	1	3.6	6.8	uF	1-4
M2	PSFC	51	2.1	3.9	uF	1-4

NOTES:

1. Power supply filter capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance bridge frequency	1 KHz	± 100 Hz	M1,M2
Capacitance bridge voltage	0.3 Vp	± 30 mVp	M1,M2
Capacitance measurement	NA NA	± 160 nF ± 90 nF	M1 M2

2. The capacitance bridge used in this test shall be item C01. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XIII.

3. The test circuit shall be as indicated on figure 10.

4. Abbreviations used are defined in 6.1.

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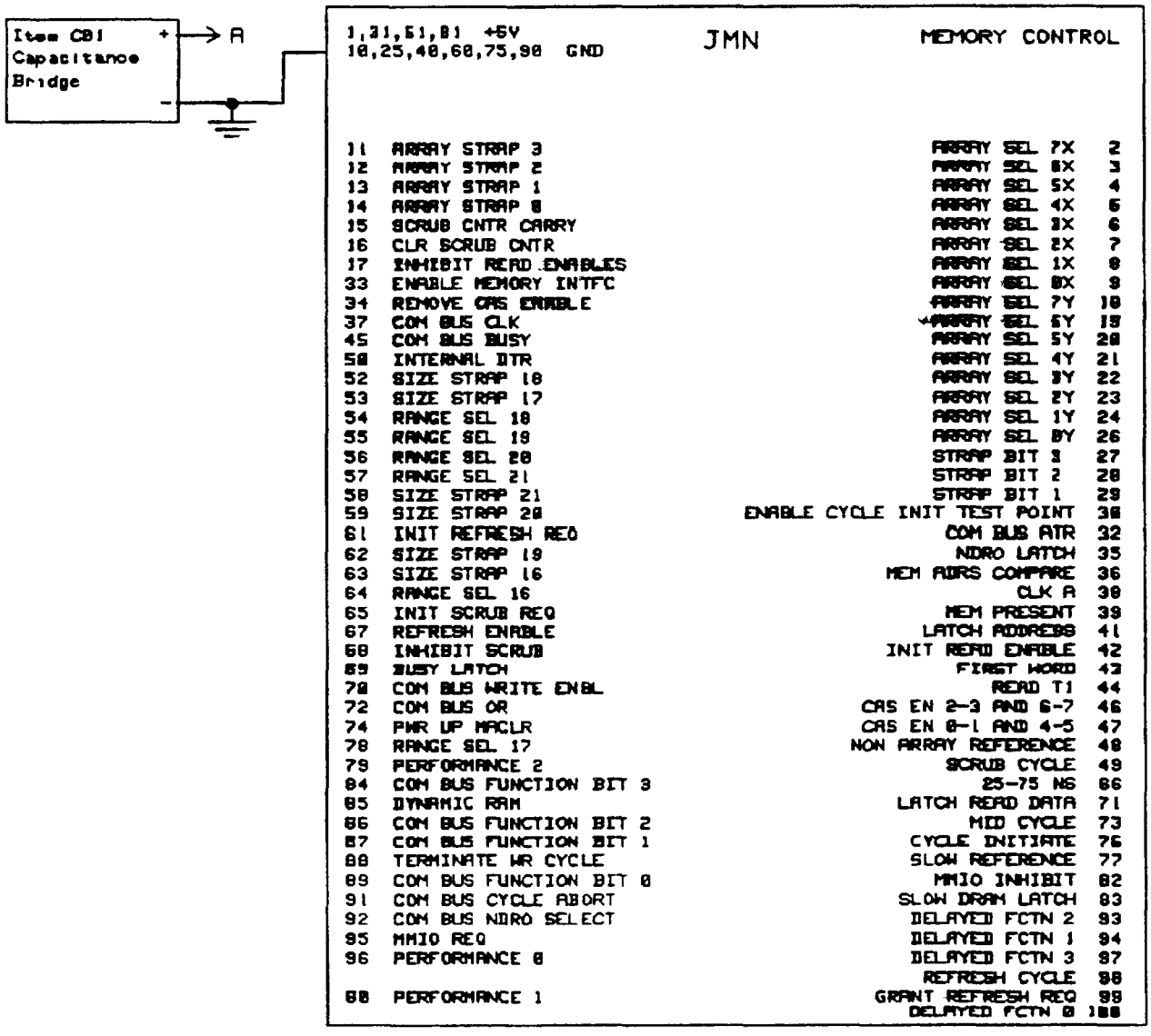


FIGURE 10. Power supply filter capacitance test circuit for table XIII.

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TABLE XIV. Power supply current test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
N	Icc	1,31 51,81	650	1560	mA	1-3

NOTES:

1. Power supply current test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.5 V	± 55 mV	N
Power supply current limit	2.34 A	NA	N
Pattern voltage high, (VH)	3.0 V	± 100 mV	N
low, (VL)	0.0 V	± 100 mV	N
Power supply current measurement	NA	± 45.5 mA	N

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Precondition the module under test and take a measurement at the pattern line with the maximum power supply current as indicated in table XIV.

3. Abbreviations used are defined in 6.1.

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TABLE XV. Functional test.

Test cond	Test description	Notes
P1	Functional test minimum power supply voltage maximum pattern rate (225 nS)	1-3
P2	Functional test maximum power supply voltage maximum pattern rate (225 nS)	1-3

NOTES:

1. Functional test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	P1 P2
Power supply current limit	2.34 A	NA	P1,P2
Pattern voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 P1,P2 P1,P2
Output voltage verification high, (Voh) high, (Voh) low, (Vol)	 NA NA NA	 ± 105 mV ± 155 mV ± 20 mV	 P1 P2 P1,P2

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NOTES: - Continued.

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

(b) Use the appropriate pattern to verify the functional performance of the module as indicated in table XV using only a capacitive load of 100 pF (see figure 9). The output compare levels shall be as follows:

High level: $2.4\ V \leq Voh \leq Vcc$
Low level: $0.0\ V \leq Vol \leq 0.5\ V$

3. Abbreviations used are defined in 6.1.

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TABLE XVI. Dynamic test.

Test cond	Test description	Notes
Q1	Dynamic test, sample maximum pattern rate (225 nS) minimum power supply	1-6
Q2	Dynamic test, sample maximum pattern rate (225 nS) maximum power supply	1-6
Q3	Dynamic test, qualification maximum pattern rate (225 nS) minimum power supply	1-6
Q4	Dynamic test, qualification maximum pattern rate (225 nS) maximum power supply	1-6

NOTES:

1. Dynamic test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	4.5 V 5.5 V	± 45 mV ± 55 mV	Q1,Q3 Q2,Q4
Power supply current limit	2.34 A	NA	Q1-Q4
Input voltage high, (VH) low, (VL)	 2.4 V 0.4 V	 ± 100 mV ± 100 mV	 Q1-Q4 Q1-Q4

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Output voltage verification			
high, (Voh)	NA	± 100 mV	Q1,Q3
high, (Voh)	NA	± 100 mV	Q2,Q4
low, (Vol)	NA	± 100 mV	Q1-Q4

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

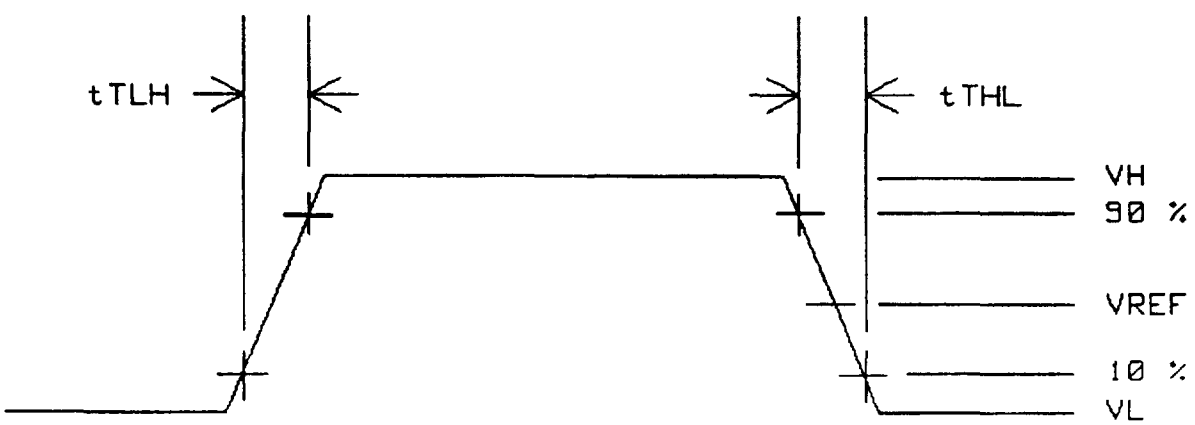
(b) The output compare levels shall be as follows:

High level: Vref ≤ Voh ≤ Vcc
Low level: 0.0 V ≤ Vol ≤ Vref

(c) The dynamic parameters are found in table XVII.

3. The load circuit for this test is specified on figure 9.

4. Input/Output (I/O) waveforms are as referenced on figure 11. All of the input waveforms shall have the following characteristics:



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NOTES: - Continued.

Where:

(a) VH and VL are as specified in the test tolerance table.

(b) tTLH = tTHL $\leq$ 10 nS.

(c) Vref = 1.6 V.

5. This test shall be performed as indicated in table XVI.

6. Abbreviations used are defined in 6.1.

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TABLE XVII. Dynamic parameters.

Ref num	Sym	Parameter	Minimum limit			Maximum limit			Unit	Figs
			-55°C	25°C	85°C	-55°C	25°C	85°C		
1	tpd	COM BUS CLK to SCRUB CYCLE	-	-	-	65	65	65	nS	11
2	tpd	COM BUS CLK to LATCH ADRESS	-	-	-	55	55	55	nS	11
3	tpd	INHIBT RD EN to NONARRAY REF	-	-	-	25	25	25	nS	11
4	tpd	COM BUS CLK to CLK A	-	-	-	14	14	14	nS	11
5	tpd	EN MEM INTFC to MEM PRESENT	-	-	-	26	26	26	nS	11
6	tpd	COM BUS CLK to CSEN 2-3&6-7	-	-	-	47	47	47	nS	11
7	tpd	PERFORMANCE 2-0 to SLOW REF	-	-	-	14	14	14	nS	11
8	tpd	COM BUS CLK to DELAYEDFCTN 3-0	-	-	-	44	44	44	nS	11
8	tpd	COM BUS CLK to SLOW DRAM LA	-	-	-	44	44	44	nS	11
9	tpd	COM BUS CLK to READ T1	-	-	-	61	61	61	nS	11
10	tpd	COM BUS CLK to REFRESH CYC	-	-	-	54	54	54	nS	11

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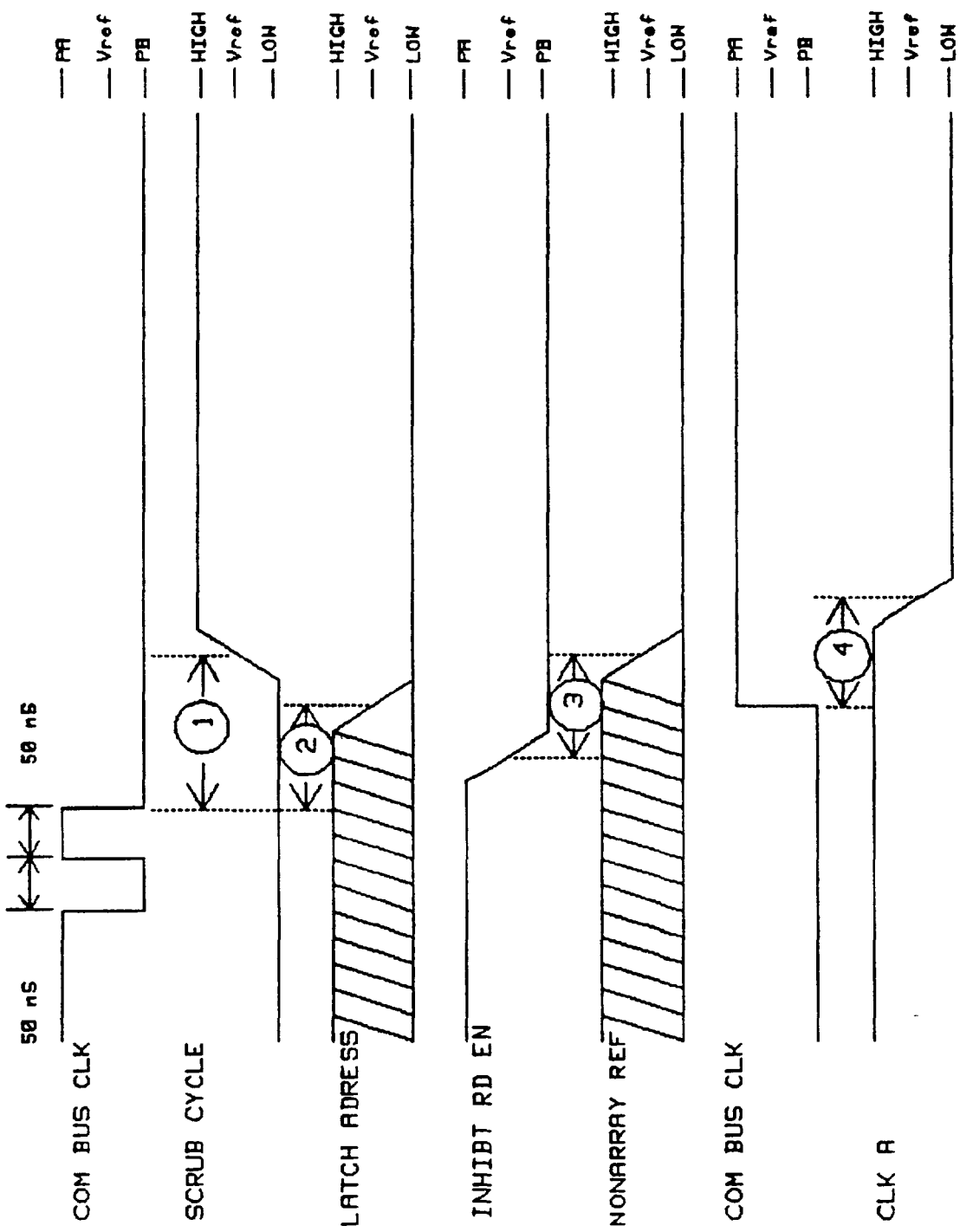


FIGURE 11a. Dynamic waveforms for table XVII.

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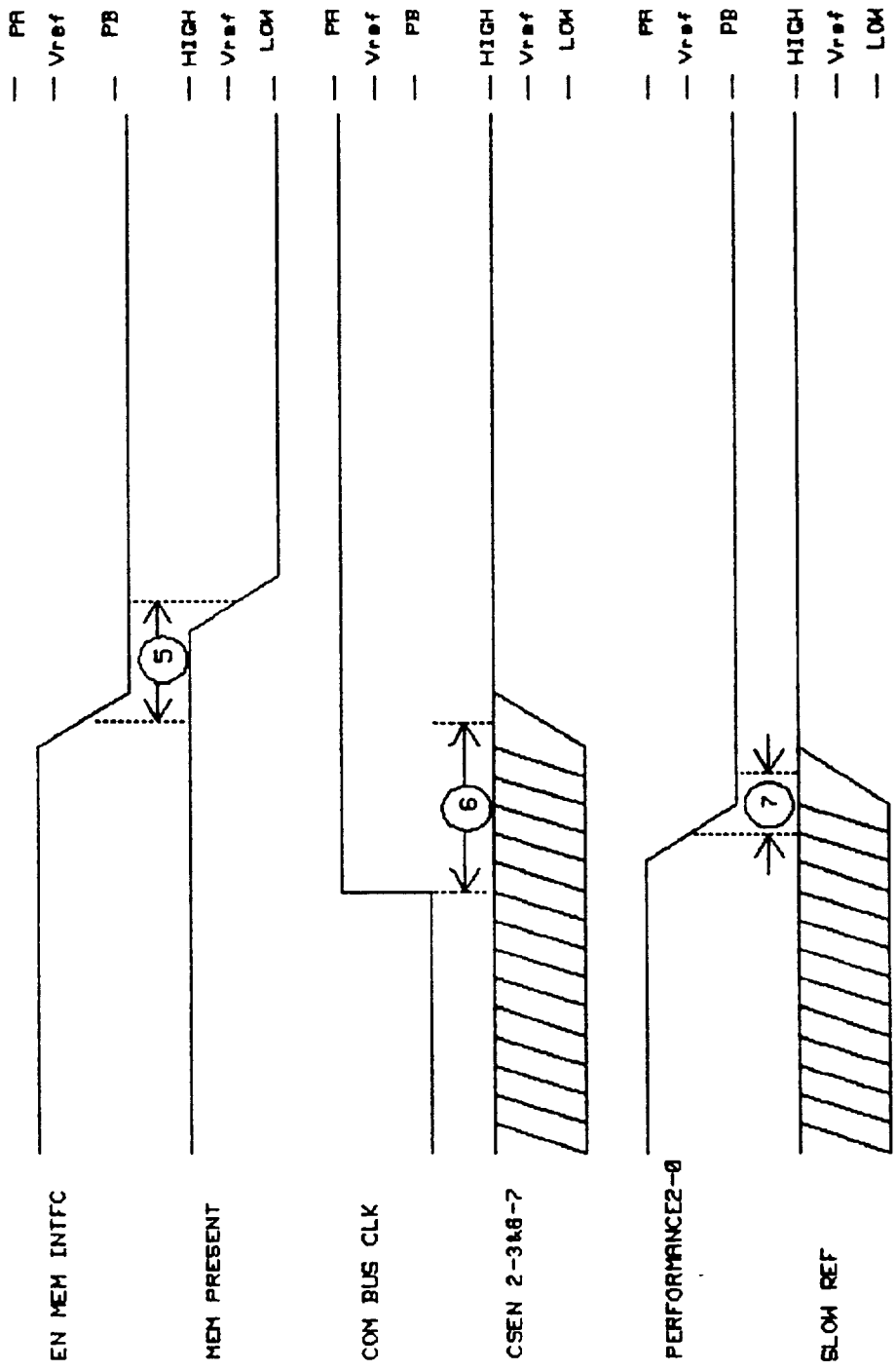


FIGURE 11b. Dynamic waveforms for table XVII. - Continued.

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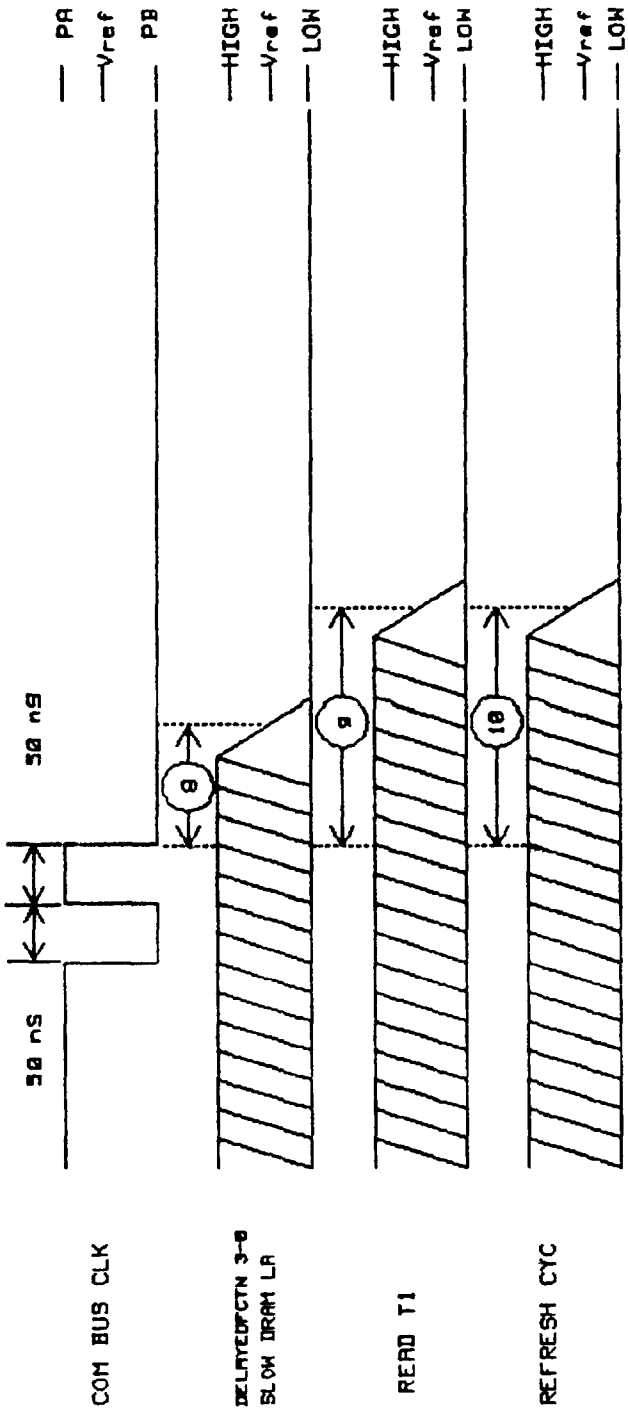


FIGURE 11c. Dynamic waveforms for table XVII. - Continued.

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TABLE XVIII. Input and output capacitance test.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
Inputs						
R1	ARRAYSTRAP 3	11	2	75	pF	1-4
R2	ARRAYSTRAP 2	12				
R3	ARRAYSTRAP 1	13				
R4	ARRAYSTRAP 0	14	2	75	pF	1-4
R5	SCRUB CTR CA	15	4	100	pF	1-4
R6	CL SCRUB CTR	16	2	75	pF	1-4
R7	INHIBT RD EN	17				
R8	EN MEM INTFC	33				
R9	RMOVE CAS EN	34	2	75	pF	1-4
R10	COM BUS CLK	37	12	100	pF	1-4
R11	COM BUS BUSY	45	2	50	pF	1-4
R12	INTERNAL DTR	50	4	75	pF	1-4
R13	SIZESTRAP 18	52	4	50	pF	1-4
R14	SIZESTRAP 17	53	2	50	pF	1-4
R15	RANGE SEL 18	54				
R16	RANGE SEL 19	55				
R17	RANGE SEL 20	56				
R18	RANGE SEL 21	57				
R19	SIZESTRAP 21	58				
R20	SIZESTRAP 20	59	2	50	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R21	IREFRESH REQ	61	2	50	pF	1-4
R22	SIZESTRAP 19	62				
R23	SIZESTRAP 16	63				
R24	RANGE SEL 16	64				
R25	ISCRUB REQ	65	2	50	pF	1-4
R26	REFRESH EN	67	4	75	pF	1-4
R27	INHIBITSCRUB	68	2	50	pF	1-4
R28	BUSY LATCH	69	4	200	pF	1-4
R29	CBUS WR EN	70	2	75	pF	1-4
R30	COM BUS OR	72	6	100	pF	1-4
R31	PWR UP MACLR	74	2	75	pF	1-4
R32	RANGE SEL 17	78	2	75	pF	1-4
R33	PERFORMANCE2	79	2	200	pF	1-4
	Bidirectional					
R34	PERFORMANCE1	80	6	100	pF	1-4
	Inputs					
R35	CBUS FCTN B3	84	4	100	pF	1-4
R36	DYNAMIC RAM	85	6	75	pF	1-4
R37	CBUS FCTN B2	86	2	50	pF	1-4
R38	CBUS FCTN B1	87	2	50	pF	1-4
R39	TERMI WR CYC	88	2	216	pF	1-4

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TABLE XVIII. Input and output capacitance test - Continued.

Test cond	Signal name	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
R40	CBUS FCTN B0	89	4	100	pF	1-4
R41	CBUS CYC ABT	91	2	50	pF	1-4
R42	CBUS NDRO SL	92	4	75	pF	1-4
R43	MMIO REQ	95	4	75	pF	1-4
R44	PERFORMANCE0	96	2	175	pF	1-4

NOTES:

1. Input and output capacitance test tolerance table.

Condition	Value	Accuracy	Test condition
Capacitance meter frequency	1 MHz	± 100 KHz	R1-R44
Capacitance meter voltage	15 mV	± 1.5 mV	R1-R44
Capacitance measurement	NA	± 3.65 pF	R1-R4
	NA	± 4.8 pF	R5
	NA	± 3.65 pF	R6-R9
	NA	± 4.4 pF	R10
	NA	± 2.4 pF	R11
	NA	± 3.55 pF	R12
	NA	± 2.3 pF	R13
	NA	± 2.4 pF	R14-R25
	NA	± 3.55 pF	R26
	NA	± 2.4 pF	R27
	NA	± 4.8 pF	R28
	NA	± 3.65 pF	R29
	NA	± 4.7 pF	R30
	NA	± 3.65 pF	R31,R32

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NOTES: - Continued.

Condition	Value	Accuracy	Test condition
Capacitance measurement	NA	± 9.9 pF	R33
	NA	± 4.7 pF	R34
	NA	± 4.8 pF	R35
	NA	± 3.45 pF	R36
	NA	± 2.4 pF	R37,R38
	NA	± 10.7 pF	R39
	NA	± 4.8 pF	R40
	NA	± 2.4 pF	R41
	NA	± 3.55 pF	R42,R43
	NA	± 8.65 pF	R44

2. The capacitance meter used in this test shall be item C05. The negative terminal shall be connected to the module ground and the positive terminal to the pin indicated in table XVIII.

3. The test circuit shall be as indicated on figure 12.

4. Abbreviations used are defined in 6.1.

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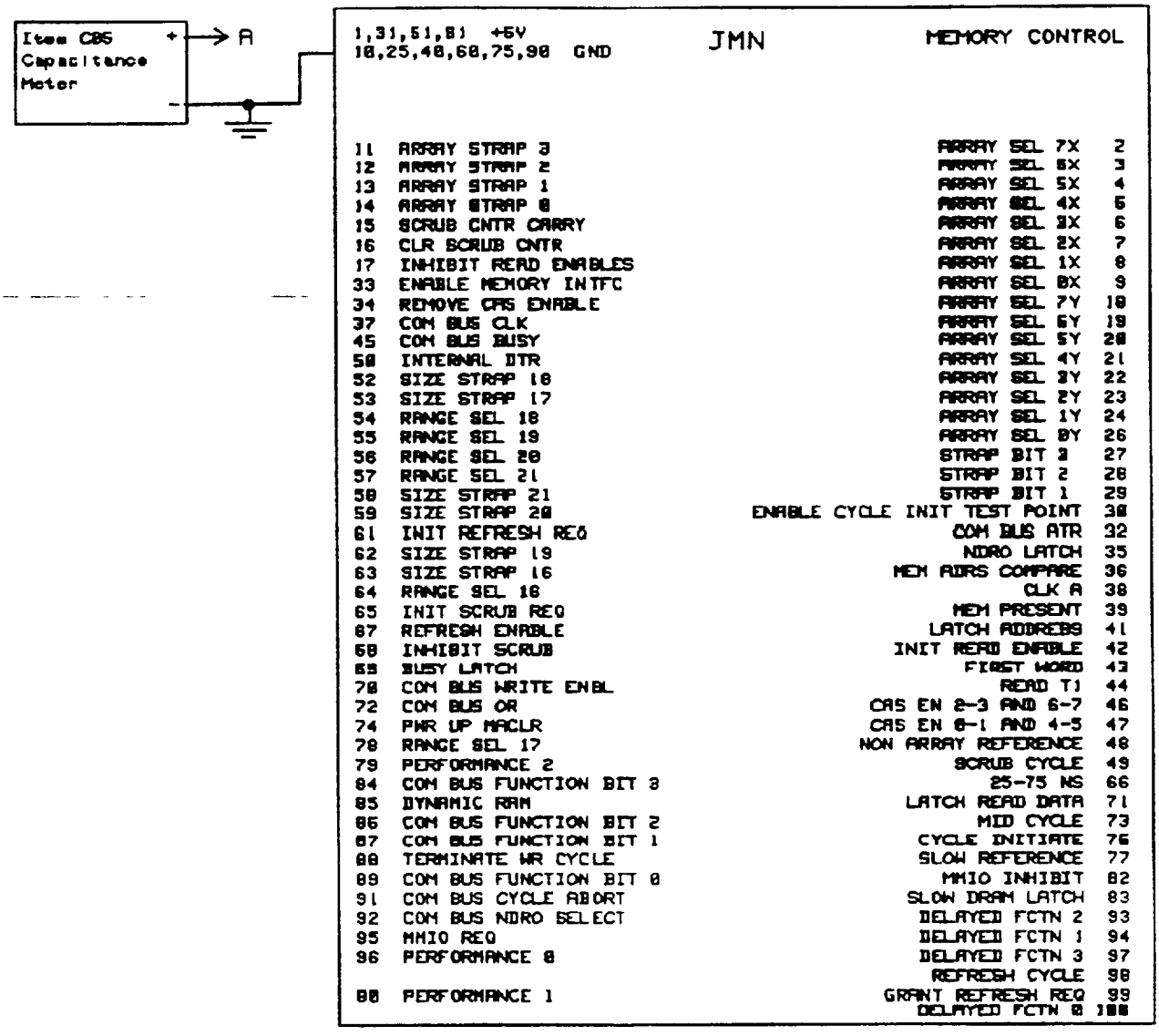


FIGURE 12. Input and output capacitance test circuit for table XVIII.

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TABLE XIX. Power supply transient amplitude test.

Test cond	Parameter	Pin no.	25 ± 5°C Initial limits Min	Max	Units	Notes
T1	PSTA	1	0	400	mV	1-4
T2	PSTA	51	0	400	mV	1-4

NOTES:

1. Power supply transient amplitude test tolerance table.

Condition	Value	Accuracy	Test condition
Power supply, (Vcc)	5.0 V	± 250 mV	T1,T2
Power supply current limit	2.34 A	NA	T1,T2
Pattern voltage high, (VH) low, (VL)	 3.0 V 0.0 V	 ± 100 mV ± 100 mV	 T1,T2 T1,T2
Power supply transient amplitude measurement	NA	± 20 mV	T1,T2

2. The following procedure is required to perform this test:

(a) The module power supply and ground pins shall be connected as specified in the power supply stress test table to an external source. The power supply voltage, including overshoot, shall not exceed the value specified in the power supply stress test tolerance table.

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NOTES: - Continued.

(b) Precondition the module under test and make the indicated measurement for the test condition shown in table XIX.

(c) Vcc pins 1 and 31 shall be shorted together, pins 51 and 81 shall be shorted together and the scope point, probe 'A', shall be connected to the pin indicated in table XIX.

(d) Using the oscilloscope, D06, inspect for the most negative spike as shown on figure 13 at the pattern line with the maximum power supply current.

(e) The maximum probe capacitance is 15 pF. There shall be no other capacitance present on Vcc to ground.

3. The test circuit shall be as indicated on figure 14.

4. Abbreviations used are defined in 6.1.

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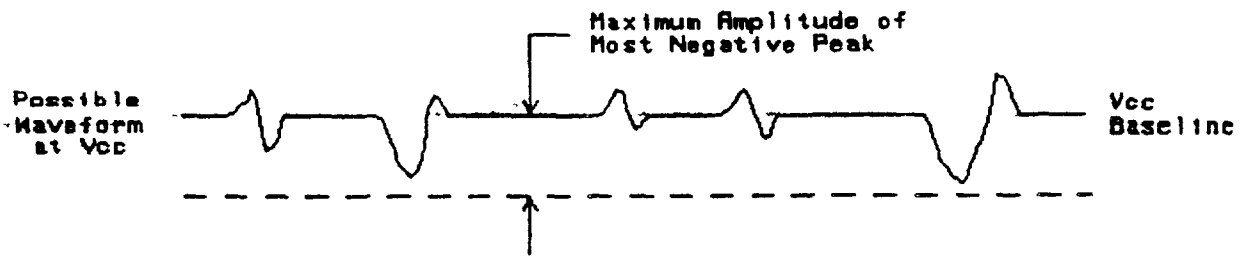
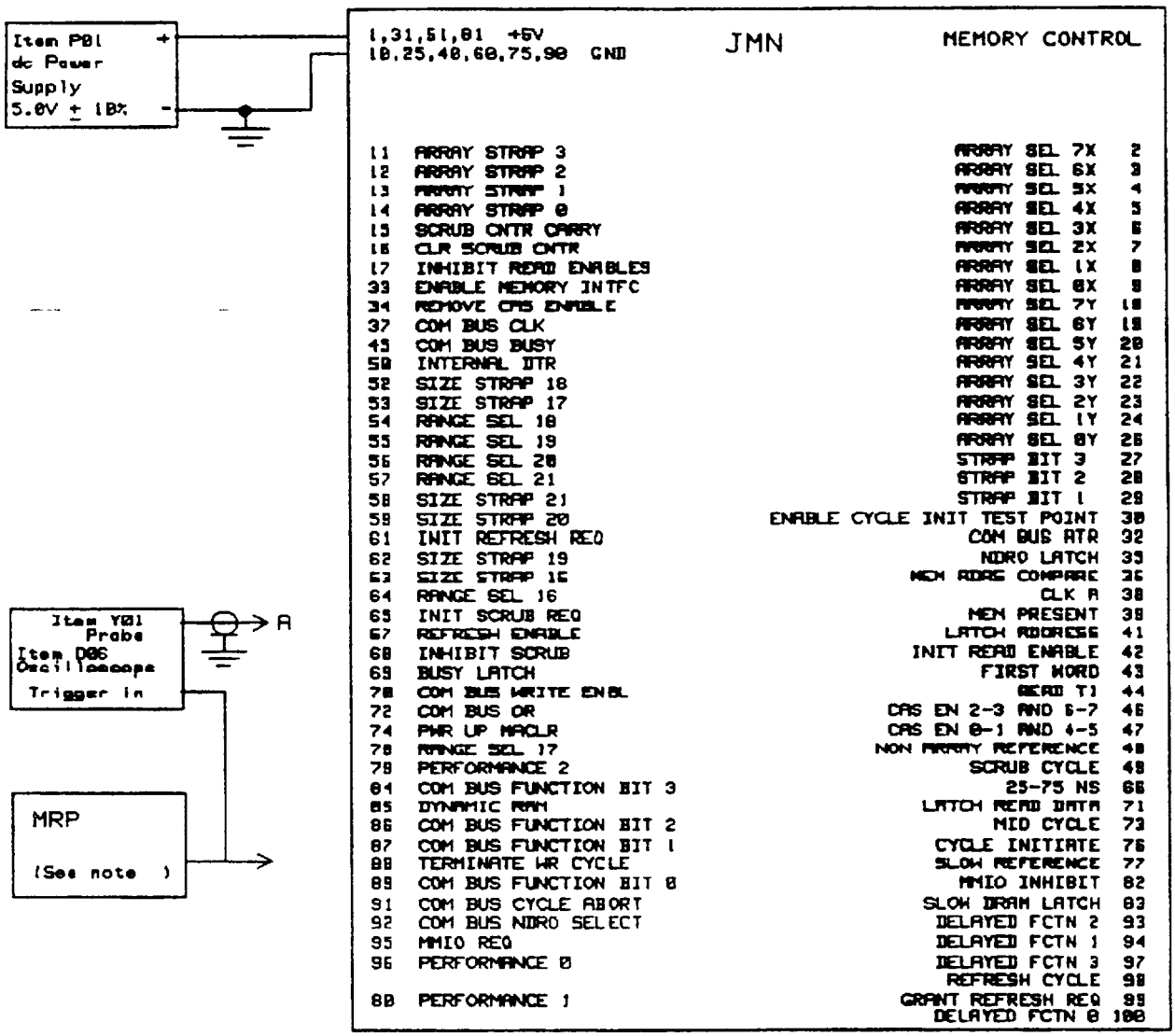


FIGURE 13. Power supply transient amplitude waveform for table XIX.

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NOTE:

1. MRP or emulator capable of reproducing system backpanel characteristics.

FIGURE 14. Power supply transient amplitude test circuit for table XIX.

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a. NAME Technical Point of Contact (TPOC): Mr. Robert F. Dilger (Code 602A) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial: TPOC: 812-854-1528 (2) AUTOVON: 88-482-1528
c. ADDRESS (Include Zip Code) Commanding Officer Naval Weapons Support Center ATTN: Robert F. Dilger, Bldg. 2940W, Code 602A Crane, Indiana 47522-5060	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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